

Mean-Variance Portfolio Rebalancing with Transaction Costs*

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Abstract

As in continuous time, the nontrading region (NTR) in a mean-variance model with fixed, proportional, and quadratic trading costs is a singleton only for pure quadratic costs. Utility loss from costs is approximately proportional at small cost levels, and approximately constant at large cost levels. Trading can be to the interior of the NTR (fixed costs with or without proportional costs), the exterior of the NTR (quadratic costs with or without proportional costs), or the boundary (pure proportional costs). Absent fixed costs, the NTR is a multidimensional parallelogram. An application shows how to improve on traditional symmetric futures overlay strategies.

JEL-Classification: C61, D23, G11

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1 Introduction

In general, optimal portfolio rebalancing given proportional and/or fixed transaction costs is a complex problem. Even with only two securities, solving for the optimal strategy in a continuous-time model involves either a primal free boundary problem¹ or its dual formulation.² With more securities, models have been solved only in the extreme case of uncorrelated returns and constant absolute risk aversion³ or with heuristic numerical approximations.⁴ To understand better the economics of transaction costs, we use a single-period mean-variance model with fixed, proportional, and quadratic costs. At small cost levels, the utility loss from transaction costs is approximately proportional to the scale of costs and roughly equal to the cost of trading to the ideal portfolio. At high cost levels, the utility loss from transaction costs is approximately constant and roughly equal to the cost of not trading. Proportional utility loss at low cost levels and constant utility loss at high cost levels should hold in dynamic models as well, except in degenerate cases. The usefulness of the analysis is demonstrated by showing the error in the assumption that a traditional futures overlay strategy should be symmetric.

Absent costs and other frictions such as taxes, the optimal portfolio after rebalancing depends on the initial portfolio only through the initial wealth, since it does not cost anything to move among positions having the same market value. However, if any of the costs we analyze are not zero, where we trade to does depend on the initial position, and it is not always optimal to trade to the ideal portfolio that would be optimal absent costs. In particular, given fixed or proportional costs (with or without quadratic costs), there exists a set of initial portfolios which are different from the ideal portfolio but too close to justify any trade, so that it is optimal not to trade at all. Locally, the constant or linear cost of

¹Davis and Norman (1990), Dumas and Luciano (1991), Liu and Loewenstein (2004), Shreve and Soner (1994) and Taksar et al. (1988)

²Goodman and Ostrov (2010) and Schachermayer (2017)

³Liu (2004)

⁴Leland (2000), Balduzzi and Lynch (1999), Balduzzi and Lynch (2000), Donohue and Yip (2003), Han (2005), Muthuraman and Kumar (2006), Irle and Prelle (2008), Lynch and Tan (2009), Myers (2009) and Buss and Dumas (2017)

moving to the ideal portfolio is a larger order of magnitude compared to the quadratic utility loss from having the wrong allocation. The set of portfolios from which it is not optimal to trade is referred to as the nontrading region (NTR). Absent costs, and in the special case of pure quadratic costs, the NTR is a single point. With proportional costs but no fixed costs, with or without quadratic costs, the nontrading region is a multidimensional parallelogram. This is consistent with the multidimensional rectangle found by Liu (2004), who assumed uncorrelated returns and exponential utility, and lends support to the assumed form of the NTR in the heuristic modeling of Leland (2000).

Many frictions can be modeled as trading costs, for example, brokerage fees, bid-ask spread, price pressure (getting a less favorable price the more you trade), time and effort to compute and submit the trade, cost of time to visit your broker or logging in to the broker's website, and the cost of studying the securities in your portfolio. The overall cost of trading is probably complex in practice, for example, there may be a fixed cost of attention, the bid-ask spread could be constant for small trades and increasing (due to price pressure) or decreasing (due to bargaining power) as trades get larger, and the brokerage fee may depend in a complicated way on the size of the trade in all the securities. In this paper we model costs as a function of the size of trade as fixed, proportional, quadratic, or a combination of these. We assume the proportional component of costs can differ across securities, and we allow the fixed cost to be any function of the subset of securities traded, with particular examples of an overall fixed cost, a per-security fixed cost, or a fixed cost for trading all securities listed on the same exchange. For quadratic costs, possibly due to price pressure, the marginal cost gets higher the more you trade, and there can be cross effects of trading securities, which we think of as being due to trading in one security affecting the price of another security (as in Back (1993)).⁵ In our view, all investors face proportional trading costs in the form of

⁵Examples of models with price pressure or quadratic costs include Leland (1985), Kyle (1985), Back (1993), and Kyle et al. (2018). The single-period version of Garleanu and Pedersen (2013) is essentially the same as our model with only quadratic costs. It's less obvious that price pressure generates something that looks like quadratic costs in continuous time. For example, in the continuous-time Kyle model of Back (1993), the impact of price pressure on the informed agent is a function of the cumulative trade, not an integral of a quadratic function of the trading rate. This is because the impact of trading on price is persistent in this sort of rational model. In a model with temporary liquidity problems, splitting trades over time might reduce

brokerage fees plus the bid-ask spread that is present even for small trades. Quadratic costs from price pressure are negligible for small and intermediate-size investors (except when taking a large position in a small stock) but important for large investors. Fixed costs are probably significant only for small investors. However, we do not impose these views in our model and instead we admit any combination of fixed, proportional, and quadratic costs. This makes our model general enough to give rich representative results yet simple enough to analyze completely.

The three different types of costs in our model are qualitatively different and have different implications for the optimal trading strategy. With only proportional costs, the marginal cost of trading in some direction is constant and independent of the size of the trade, so the marginal choice of how far to trade in some direction is the same as if you started at the point under consideration. This implies that any trade must go to the boundary of the NTR but no further.⁶ With only an overall fixed cost, if you trade at all and incur the fixed cost, it is optimal to trade to the ideal point. More generally, with fixed costs (with or without proportional costs), the marginal cost of trading in some direction that has a fixed cost is a limiting case of decreasing marginal cost (infinite for the initial trade and constant thereafter), and consequently any (nonzero) optimal trade goes inside the NTR. Starting at the boundary of the NTR, we would be just indifferent to trading, but the marginal cost is less if we started outside so we keep going. With quadratic costs (with or without proportional costs), if the quadratic cost is not zero in some direction, the marginal cost of trading in that direction is increasing in the size of the trade, so we never trade to the NTR. Interestingly, if we have increasing marginal costs (quadratic costs with or without proportional costs but without any fixed costs), the NTR is the same as it would be without the quadratic costs because, at the initial portfolio, the marginal cost of trading is unaffected

the impact of price pressure a little, but a time-separable quadratic cost function would seem to overstate by a large amount the benefit of splitting trades over time.

⁶Masters (2003) performs a mean-variance-style analysis with a single risky security and proportional trading costs in which it is claimed that it is not optimal to trade to the boundary of the NTR. However, this is because the paper computed the NTR incorrectly as the set of portfolios from which it would be worth trading to the ideal portfolio that would be chosen absent costs.

by the quadratic cost term. If we have both quadratic costs and fixed costs (with or without proportional costs), the marginal cost is neither increasing nor decreasing, and an optimal trade can end up outside, inside, or on the boundary of the NTR.

Our paper clarifies the claim by Constantinides (1986) that the impact of transaction costs is second-order because the first-order reaction to transaction costs is to trade less, and the counterclaims by Dai et al. (2016) that the impact is first-order if there are market closures and Jang et al. (2007) that the impact is first-order if there are regime switches in the investment opportunity set. We look at utility loss compared with the optimum in a model with no costs.⁷ In our model, if we do not start at the ideal portfolio (that would be optimal absent costs), the impact of transaction costs is approximately proportional for small enough costs and approximately constant for large enough costs. This is true for whether the costs are fixed, proportional, quadratic, or a combination. The result follows because when the cost level is very small, we trade approximately to the ideal portfolio that would be optimal absent costs, for which trade the utility loss is proportional to the cost scale, and the difference in utility between trading to the optimal portfolio and the ideal portfolio is second order. In a continuous-time model, if we start away from the ideal portfolio at the beginning, we would trade near there when costs are small, incurring a linear cost quite generally. Starting at the ideal portfolio is the only case in which the cost could have a second-order impact on utility as described by Constantinides (1986): the cost in our model is zero, but in a continuous-time model the desirability of subsequent trading (because of the change of position due to asset price changes or information arrival) could imply a second-order impact of cost. For Dai et al. (2016), even if we start at the ideal portfolio, market moves during nontrading periods (like predictable jumps), or in the case of Jang et al. (2007) a regime switch in the parameters, can move us outside the nontrading region (or move the nontrading region away) and there is a proportional cost to get to the NTR boundary. Therefore, in these cases the utility loss includes a proportional cost to return to

⁷Constantinides (1986) consider the “liquidity premium,” an expected return required compared to the same asset with no costs. Since our mean-variance utility is measured in units of return, our measure has the same units, but we do not have to assume there is another similar asset trading without costs.

the NTR even if we start at the ideal portfolio. In general, it seems like proportional utility loss at low cost levels is the normal case.

Alternatively, when the cost is very large, we trade approximately nothing, for which the utility loss is constant, which is the extreme version of Constantinides' trading less in reaction to costs. The result in our model that the utility loss is roughly constant at large cost levels should also be true in multiperiod models, providing it is feasible not to trade at all.

One interesting application of the model is to futures overlays. In a futures overlay, reallocation to asset classes is done in part by using futures instead of trading the underlying. For example, consider a portfolio that has two asset classes, equities and fixed-income, and has a target allocation of 60% in equities. As the market rises and falls, the allocation to equities goes up and down because equities respond much more to market changes than bonds.⁸ A plan sponsor sensitive to trading costs might reallocate between asset classes only when the allocation to equities increases to 70% or falls to 50%. With a futures overlay, the plan sponsors can take advantage of the lower trading costs for futures and do part of the rebalancing using futures instead of stocks, perhaps starting to trade when the exposure to equities (from equities and futures) increases to 65% or falls to 55%. Traditionally, a futures overlay strategy would be done symmetrically, trading in futures until the position is say 5% of the portfolio long or short, and then switching into trading the underlying if more trading is needed. Our analysis shows that if there is any reason (probably superior alpha from active management) to hold equities instead of synthetic equity (futures + bonds) in the first place, then the optimal strategy is asymmetric, using futures to sell equity exposure but buying underlying to buy equity exposure, for a large range of positions.

In models featuring both proportional costs and an overall fixed cost, if the starting portfolio is far enough from the ideal portfolio, the optimal trade is exactly the same as the optimal trade with proportional costs only. This is because once it is obvious it will be

⁸The need to reallocate after market moves is probably the main motivation for rebalancing in practice. The dynamic model of Garleanu and Pedersen (2013) is probably not useful in practice because this channel is absent: contrary to practice, the change in asset holdings is assumed to come only from rebalancing.

optimal to trade something and incur the fixed cost, we can view the fixed cost as sunk. The argument is slightly more complicated in the presence of asset-specific (or more general) fixed costs. Consider an example with independent returns and asset-specific fixed costs for each of two securities. If we are far away from the ideal position only in one security, the fixed cost for this security can be considered as sunk but the same is not true for the other security. This results in an optimal trade involving only the most displaced asset. However, in the absence of asset-specific fixed costs it would have been optimal to trade in both securities.

Absent fixed costs, our model is a quadratic program which is solved very quickly by standard software, and the solution is unique. Adding fixed costs, we have the best solution of a finite set of quadratic programs and the solution is still very fast if there are not too many different fixed costs. For existence (and uniqueness absent fixed costs), sufficient conditions are (i) the covariance matrix of returns is positive definite, (ii) the risk aversion is positive (iii) any tracking error penalty of deviating from a benchmark is nonnegative, (iii) the price pressure cost matrix is positive semi-definite, and (iv) fixed costs are nondecreasing in the set of securities traded. Condition (iv) is needed to avoid a closure problem: without condition (iv) it might be that the limit of trading smaller and smaller amounts of a security is preferred to trading none (which incurs a larger fixed costs). If condition (iv) holds, the cost of trading none is at least as good as the value in the limit, which is not a problem for existence of an optimum.

Mean-variance analysis was originated by Markowitz (1952), who described the basic formulations and the quadratic programming tools used to solve them. The theory was further described by Tobin (1958), who focused on macroeconomic implications of the theory. Early discussions of transaction costs often focused on the intuition that small investors who face high costs will choose a smaller and less diversified portfolio than will a large investor with smaller costs. This intuition has been formalized by a constraint on the number of

securities in the portfolio,⁹ a fixed cost for each security included in the portfolio,¹⁰ or a study of benefits of adding securities without modeling the costs¹¹. Aspects of our analysis were foreshadowed by Pogue (1970), who formulated but did not solve models with variable costs and other institutional features.

Although our single-period model is primarily intended to clarify the economics, Maurer et al. (2020) show it is useful for dynamic trading in FX markets: taking into account costs while optimizing over FX carry trades leads to an economically large and statistically significant improvement in the out-of-sample performance with Sharpe ratio increments of at least 25%. They also show that at least 40% of the improvement is due to the proper treatment of correlations. Another attractive empirical feature of our setup is the ease at which it scales up with the number of securities: Pezzo et al. (2020) construct profitable out-of-sample trading strategies in the U.S. stock market involving thousands of stocks.

The rest of this paper is structured as follows: Section 2 introduces the general framework, Section 3 conveys the main economic insights through the graphical analysis of a set of selected examples, Section 4 provides a formal characterization of the solutions and their properties, and Section 5 concludes.

2 The mean-variance framework

There are N risky securities and a riskless asset (cash). We could write down a model in which different activities represent many types of trades: buying a risky asset using cash, selling to convert a risky asset to cash, swapping two risky securities, or trading bundles of securities for cash. To keep the notation simple, we will assume all trades go through cash and there are no wash sales (simultaneous sale and purchase of the same stock), but the entire analysis could be generalized to include these cases. In the particular case of wash sales, either the round-trip cost is zero, in which wash sales are uninteresting and the

⁹Jacob (1974))

¹⁰Brennan (1975), Goldstein (1979), and Mayshar (1981)

¹¹Mao (1970)

availability of wash sales do not change payoffs, or the round-trip cost is positive, in which case wash sales are never optimal and still do not affect payoffs. Preferences are given by a utility function that is linear in mean and variance of end-of-period wealth. Transaction costs can be proportional, fixed as a function of the set of risky securities traded, quadratic, or any combination of these different types. We solve the following problem:

Problem 1 *Given the vector θ_0 of initial risky security positions, choose a vector of security positions to maximize the utility of terminal wealth*

$$\begin{aligned}
 U(\theta) = & \theta' \mu - \frac{\lambda}{2} \theta' \mathbf{V} \theta - \frac{\kappa}{2} (\theta - \theta_B)' \mathbf{V} (\theta - \theta_B) - C'_+(\theta - \theta_0)^+ - C'_-(\theta_0 - \theta)^+ \\
 & - K(\{n | (\theta - \theta_0)_n \neq 0\}) - \frac{1}{2} (\theta - \theta_0)' \mathbf{\Pi} (\theta - \theta_0).
 \end{aligned} \tag{1}$$

We use the following notation:

θ : *vector of risky security positions after trades have occurred*

μ : *vector of risky security expected excess returns net of any liquidation costs*

λ : *risk aversion parameter (typically positive)*

κ : *tracking error parameter (typically zero or positive)*

$\mathbf{V}$: *positive semi-definite covariance matrix of risky security returns net of any liquidation costs*

θ_B : *vector of benchmark risky security positions*

$C_+ \geq 0$: *vector of proportional transaction cost parameters for purchases*

$C_- \geq 0$: *vector of proportional transaction cost parameters for sales*

x^+ : *the componentwise positive part of x , $(x^+)_n = \max(x_n, 0)$*

$(\theta - \theta_0)^+$: *the vector of purchases to rebalance from θ_0 to θ*

$(\theta_0 - \theta)^+$: *the vector of sales to rebalance from θ_0 to θ*

$\mathcal{S}$: *the set of all tradable risky securities*

$\{n | (\theta - \theta_0)_n \neq 0\}$: *the set of risky securities actually traded*

$K(S)$: *fixed cost for trading in subset S , $(\forall S) K(S) \geq 0, K(\emptyset) = 0$, nondecreasing as we make the subset larger i.e. $(\forall S_1 \subseteq S_2 \subseteq \mathcal{S}) (K(S_1) \leq K(S_2))$*

$\mathbf{\Pi}$: matrix of price pressure parameters (positive semi-definite, $\mathbf{V} + \mathbf{\Pi}$ positive definite)

The vector of risky security positions after trade is θ , and the riskless-asset position is implicitly given as a residual, so that the positions in θ do not have to add up to the initial wealth. Utility is given as an excess over what we would get from investing only in the riskless asset without costs. We are suppressing the prices of the securities, which can be thought of as exogenous or perhaps reflected implicitly in the case of a quadratic cost we think of as due to price pressure. If the securities are assets such as stocks or bonds, we can think of θ as being the number of dollars invested at the current price, and μ as the vector of expected rates of return in excess of the riskfree rate. In the case of futures, we enter at a price of zero and we can think of θ being measured in terms of the size of the position (in dollars worth of the underlying), with the excess return and volatility being measured at that scale. The excess return to a futures contract is the expected change in futures price because a futures contract is “already levered.” It requires no investment, or we can say that the interest we subtract off is the riskfree rate times the initial investment, which is 0.

The first two terms in the utility function (1) are standard for mean-variance optimization: $\theta'\mu$ gives the net change in expected return from holding risky securities rather than just the riskless asset, while $\frac{\lambda}{2}\theta'\mathbf{V}\theta$ is the utility penalty for taking on variance $\theta'\mathbf{V}\theta$. The constant $\lambda > 0$ is the coefficient of risk aversion: the larger the value of λ , the more reluctant we are to take on risk in exchange for return.¹² Our model is theoretical and we are mostly ducking issues of empirical estimation, but we should emphasize that using this model in practice requires reasonable forward-looking measures of the mean and variance, and it has been known for a long time, the historical sample statistics are not reasonable.¹³

¹²Including 2 in the denominator makes the units the same as absolute risk aversion in a multivariate normal model with exponential utility and initial wealth normalized to 1. Also the 2 cancels when we look at the first-order conditions.

¹³In particular, sample covariance matrices tend to be singular or nearly singular, and sample means tend to be dominated by statistical error. Near singularity of the covariance matrix plus noisy sample means imply unreasonable extreme portfolio weights. A traditional approach to dealing with this uses Bayesian analysis (e.g. Bawa et al. (1979) or Harvey and Zhou (1990)). Recently, it is more common to use *ad hoc* “big data” style tools such as average regression, variants of LASSO, or machine learning to form estimates (e.g. Gu et al. (2020)). Bayesian analysis (e.g. with Monte Carlo techniques) can easily combine estimation and optimization. Olivares-Nadal and DeMiguel (2016) have emphasized the superficial similarity between

The third term $\frac{\kappa}{2}(\theta - \theta_B)' \mathbf{V}(\theta - \theta_B)$ is a penalty for tracking error, as in Grinold and Kahn (1995). This term is perhaps controversial (and many academics would like to set $\kappa = 0$) because it depends on the benchmark θ_B and not just on the distribution of returns. The dependence on the benchmark might be unnecessary and damaging in an ideal world, but does arise in practice for good reasons and should be very familiar to practitioners. When plan sponsors allocate funds to different managers the penalty for tracking error gives the managers an incentive not to deviate too much from the plan sponsor's planned allocation reflected in managers' benchmarks. The plan sponsor may care about preserving the intended allocation because it is diversifying or because it reflects the plan sponsor's view of which asset classes are likely to do well. Alternatively, for asset-liability management, the benchmark can be interpreted as the best projection into the space of traded securities of some liability claim that a firm would like to cost-efficiently hedge by trading according to the mean-variance criterion.

The remaining terms are the costs, in future value, paid to re-balance the initial position θ_0 to θ . The vector $C_+ \geq 0$ (resp. $C_- \geq 0$) is the security-specific proportional cost of buying (resp. selling) the security. Buying $(\theta - \theta_0)_n^+$ (resp. selling $(\theta_0 - \theta)_n^+$) incurs a cost of $C_{+,n}(\theta - \theta_0)_n^+$ (resp. $(\theta_0 - \theta)_n C_{-,n}$).

The fixed cost if we trade the risky securities in $S \subseteq \mathcal{S}$ is $K(S) \geq 0$. The set of securities we trade is given by $\{n | (\theta - \theta_0)_n \neq 0\}$. In the absence of trading, no cost is incurred, so we set $K(\emptyset) = 0$. Also, we assume it costs no less to trade more securities: $(\forall S_1 \subseteq S_2 \subseteq \mathcal{S})(K(S_1) \leq K(S_2))$. If $K(S)$ is the same positive constant for all nonempty S , $\emptyset \subset S \subseteq \mathcal{S}$, that constant is an overall fixed cost. When the fixed cost function $K(\cdot)$ has the form $K(S) = \sum_{n \in S} K(\{n\})$, we have pure security-specific fixed costs. As a third example, if $\mathcal{S}$ is partitioned into M different disjoint "markets" $S_1, S_2, \dots, S_M$ with $\cup_{m=1}^M S_m = \mathcal{S}$, suppose we can write $K(S) = \sum_{m \in T} K(S_m)$ where $T \equiv \{m \in M | S \cap S_m \neq \emptyset\}$. Then we can say that we have pure market-specific fixed costs. If there are no fixed costs, then

proportional costs and a LASSO-style penalty term, but this is not a good way to look at things in general. For example, using only an *ad hoc* tuning parameter (as in LASSO and its variants) would fail to capture the cost difference between trading futures and trading equities in our futures overlay example.

$$(\forall S)(K(S) = 0).$$

Finally, the price pressure cost $-\frac{1}{2}(\theta - \theta_0)' \mathbf{\Pi}(\theta - \theta_0)$ is quadratic in the size of the net trades $\theta - \theta_0$. The diagonal elements of $\mathbf{\Pi}$ describe the security-specific price impact parameters while (possibly) non-zero off diagonal elements admit that (as in Back (1993)) trading in one security may impact the price in other securities (because learning about them is correlated).¹⁴ We assume that $\mathbf{\Pi}$ is positive semi-definite so that trading more in some direction never costs less. We think of $\mathbf{\Pi} = 0$ as applying to investors whose trades are not large enough to have a significant impact on prices. In the absence of any costs (i.e. with $C_+ = C_- = 0$, $(\forall S)(K(S) = 0)$, and $\mathbf{\Pi} = 0$), Problem 1 reduces to a traditional mean-variance problem with the penalty for tracking error as in Grinold and Kahn (1995).

Before turning to some numerical examples, we note sufficient conditions for existence and uniqueness of a solution to Problem 1.

Theorem 1 *Assume that the $(\lambda + \kappa)V + \mathbf{\Pi}$ is positive definite, and that fixed costs are nondecreasing in the set of risky securities traded, $(\forall S_1, S_2 \subseteq \mathcal{S})(S_1 \supseteq S_2 \Rightarrow K(S_1) \geq K(S_2))$. Then*

1. *If there are no fixed costs, $(\forall S)K(S) = 0$, then Problem 1 has a unique optimal solution.*
2. *In the presence of fixed costs, Problem 1 has a solution, but it may not be unique.*

Proof. Consider first case 1 without fixed costs. Since $(\lambda + \kappa)V + \mathbf{\Pi}$ is positive definite, the quadratic terms of $U(\theta)$ (for variance, tracking error, and price pressure) are strictly concave (with Hessian $-(\lambda + \kappa)V - \mathbf{\Pi}$) the proportional cost terms are concave, and the remaining terms (from expanding $\frac{1}{2}(\theta - \theta_0)' \mathbf{\Pi}(\theta - \theta_0)$) are constant or linear, so $U(\theta)$ is strictly concave. By dominance, the optimum for all of $\mathfrak{R}^N$, if any, is the same as the optimum over the set $\{\theta \in \mathfrak{R}^N | U(\theta) \geq U(\theta_0)\}$, and is also the optimum over the weakly

¹⁴The equilibrium in Back (1993) has learning about one asset from the order flow of the other because he studied a stock and an option on the stock, which are based on the same underlying value. However, in equilibrium the two are not locally perfectly correlated because the local variance changes based on the order flow. While that model has only two risky securities, this feature is general.

larger set $B \equiv \{\theta | \theta' \mu - \frac{\lambda}{2} \theta' \mathbf{V} \theta - \frac{\kappa}{2} (\theta - \theta_B)' \mathbf{V} (\theta - \theta_B) - \frac{1}{2} (\theta - \theta_0)' \mathbf{\Pi} (\theta - \theta_0) \geq U(\theta_0)\}$ in which we left out the proportional cost in $U(\theta)$. The set B is compact and nonempty (as proven in Lemma 1): because the quadratic function in the definition has a negative-definite Hessian, B is the some-dimensional analogue of a filled ellipse (an interval, an ellipse, an ellipsoid, or a hyperellipsoid). Since a continuous function on a compact set has an optimum, Problem 1 has a solution, and because U is strictly concave, the optimal value of θ is unique.

Now consider case 2, which is more subtle because the objective function is no longer continuous. For each subset S of the securities, consider Problem 1 except with two changes: (1) we impose $(\theta - \theta_0)_n = 0$ for all $n \notin S$ and (2) we set $K = K(S)$ even if we do not trade all securities in S . Then this problem has the same form as in the case 1, but for the same (if $S = \mathcal{S}$) or a smaller set of risky securities and therefore has a unique solution (since $(\lambda + \kappa)\mathbf{V} + \mathbf{\Pi}$ positive definite over all the securities implies it is positive definite over subsets). Any of the best of these finitely many solutions for all subsets S is the solution for the problem, unless each of the best solutions uses a smaller subset S' of the securities than the subset S for which it is intended. This case seems to have a closure problem, since nearby points have the correct subset and almost the same value, but not the point we are considering. However, this cannot happen given our assumption that $K(S)$ is nondecreasing, which implies that the cost is no larger (and therefore the value is no smaller) if we look at the value for the subset $S' \subset S$ for which the solution is still feasible and the cost $K(S') \leq K(S)$. To show the solution may not be unique, consider any example in which adding securities would be valuable absent fixed costs ($K(S) \equiv 0$) and change $K(S)$ to just offset the change in value from adding securities. ■

If there are not any fixed costs, the cost function is convex (marginal cost is positive in all directions), so that Problem 1 is a convex optimization problem with a unique solution. If there are fixed costs, the cost function is not convex (since the marginal cost is decreasing when we go from no trade to a positive or negative trade), but conditional on the set of securities traded the problem is still a convex optimization, and we can view Problem 1 as a kind of combinatorial problem of looking for the optimal subset to trade, with a simple convex

optimization conditional on the subset. The assumption that the fixed cost is nonincreasing in the set of securities is required to avoid a closure problem. For example, suppose that trading in the second asset does not improve our risk or return over just trading the first asset, but trading in both securities decreases the fixed cost: $K(\{1, 2\}) < K(\{2\})$, and that the roundtrip cost of trading security 2 is not zero. Then what we would like to do is to trade just the first asset but have the lower cost of trading both securities. We can get this only in the limit of trading smaller and smaller amounts of the second asset, which is costly except in the limit. This is the closure problem that is no longer present if we make the reasonable assumption of nondecreasing fixed costs.

Corollary 1 *In Theorem 1, the condition that $(\lambda + \kappa)V + \Pi$ is positive definite can be replaced by any one of the following sufficient conditions:*

1. V positive definite, $\lambda > 0$, $\kappa \geq 0$, and Π positive semi-definite
2. V positive definite, $\lambda = 0$, $\kappa > 0$, and Π positive semi-definite
3. V positive semi-definite, $\lambda + \kappa \geq 0$, and Π positive definite

Proof. Immediate from Theorem 1 and the definitions of positive definite and positive semi-definite matrices. ■

In Theorem 1, if $(\lambda + \kappa)V + \Pi$ is only positive semi-definite but not positive definite, there may still be an optimum but some structure has to be imposed to rule out arbitrage in some direction η in which $\eta'((\lambda + \kappa)V + \Pi)\eta = 0$. In Corollary 1, the first case is probably the most common case: the covariance matrix is positive-definite and we care about variance and possibly about deviation from the benchmark and price pressure. In the second case, the covariance matrix is positive-definite and we care about tracking error but not variance per se. In the third case, we may not be risk averse and we may not care about tracking error, but all trades have price pressure. This is the case in many Kyle models with risk-neutral traders.

3 Examples

In this section we illustrate and motivate the scope of our framework via a set of selected examples.

3.1 Single risky security and the utility loss from costs

Figure 1 shows how the portfolio choice and utility loss compared to the optimal trade absent costs varies as a function of the scale of costs, which is a multiplier for the whole cost function. In the example, we start with a portfolio $\theta_0 = 0.5$ and the ideal portfolio $\theta_I = 0.625$. In all cases, the optimal portfolio starts at the ideal portfolio and tends towards no trade as the cost level rises (with convergence at finite cost except in the case of quadratic costs). For pure proportional costs, the optimal portfolio starts at the ideal portfolio and moves linearly towards the initial portfolio and stays there once it reaches it. Beyond that level of cost, the cost of trading at the margin exceeds the benefit. For pure fixed cost, the optimal portfolio is the ideal portfolio until the cost increases to where it is optimal not to trade at all. With pure fixed cost, once you are trading, the marginal cost of trading is zero, so if you are trading it is best to trade all the way to the ideal portfolio. For pure quadratic costs, the optimal portfolio choice starts at the ideal point and follows a hyperbola that converges to the initial choice as the cost increases without bound. In the mixed example with all three types of fixed cost, the optimal portfolio starts at the ideal point and follows a hyperbola until the point where it is no longer worth incurring the fixed cost so the portfolio jumps to the initial portfolio.

Constantinides (1986) has argued that the impact of transaction costs is second-order, while Dai et al. (2016) and Jang et al. (2007) have argued that the impact is first-order for some deviations from the original model. The utility loss in the second graph in Figure 1 illustrates the results derived in Subsection 4.3 for the general case that may have many securities. The main result shows the asymptotic shape of the cost for small and large costs. For small costs, the utility loss is roughly proportional, approximately equal to the utility

loss from trading to the ideal point, which is first-order unless it is always costless to trade to the ideal point. For large costs, the utility loss is approximately constant and equal to the cost of not trading. These results are true for all types of cost: proportional, fixed, quadratic, or any combination of the three. In a multiperiod model, the results would be the same except when trading to the ideal point is costless.¹⁵ If it is costless to trade to the ideal point, for example if we start at the ideal point, the cost in a multiperiod model could still be positive because price movements or information arrival can move us away from the ideal point. This is consistent with Constantinides (1986), but we will get instead the result of Dai et al. (2016) or Jang et al. (2007) if there is any possibility that lumpy information arrival or a jump in prices will take us out of the nontrading region. If we start away from the ideal portfolio, the initial trade to the nontrading region will incur proportional costs as in the single-period model, so the utility loss from transaction costs will be proportional, not quadratic. Approximately constant utility loss at large cost levels will also be robust to an intertemporal model, provided it is feasible to refrain from ever trading.

To summarize, we find the utility loss at small cost is proportional (as in Dai et al. (2016) and Jang et al. (2007)), and that should hold in multiple periods absent some very special assumptions (we start at the ideal point and we have no jumps in prices or information arrival that would take us outside the nontrading region). This result is true not only for the proportional transaction costs studied by Constantinides (1986), Dai et al. (2016) and Jang et al. (2007), but also for fixed and quadratic costs, as well as hybrid cases. We also have a new result that at large costs, the utility loss from transaction costs converges to a constant, equal to the utility loss from not trading at all.

¹⁵For a reasonable model of price pressure, at least one component of cost should be based on the cumulative trading as in the continuous-time Kyle model of Back (1993), not as a time-separable quadratic function of the trading rate. For such a model, consequently, spreading the trades over time is not effective in reducing the order of price pressure.

3.2 Proportional costs

Figure 2 shows a representative example with pure proportional costs: $C_+ \gg 0$, $C_- \gg 0$, $\lambda > 0$ (risk averse), $\kappa = 0$ (no penalty for tracking error), $K(S) \equiv 0$ (no fixed costs), $\Pi = 0$ (no price pressure), and V positive definite (no redundant assets). If the initial allocation θ_0 is in the NTR, the yellow region labeled “no trade”, then there is no trade whose benefit covers the cost and it is best not to trade. The right boundary of the NTR is part of the line along which we are just indifferent at the margin about selling security 1, and it is optimal to sell security 1 if we start to the right of this boundary. The left boundary of the NTR is part of the line along which we are just indifferent at the margin about purchasing security 1, and it is optimal to purchase security 1 to the left of this boundary. The boundaries for purchasing and selling are different because the costs put a wedge between the effective price with and without costs. Similar to the case of security 1, we sell security 2 if we start above the top boundary and we buy security 2 if we start below the bottom boundary. If we start inside one of the four corner regions, then we trade in both securities. Absent the positive correlations between the returns of the two securities, the NTR would have been a square (or a rectangle for non-identical costs) with sides parallel to the axes. With positive correlation (or a positive weight on benchmark deviations and positive correlations with the benchmark), the two securities are substitutes, and over-weighting in one security is more serious if we also have over-weighting in the other security. This is why the NTR is larger along the -45° direction in which the over- and under-weightings cancel than along the 45° direction in which the over- and under-weightings reinforce each other.

This example assumes $\Pi = 0$ so we do not have a “price pressure” quadratic term. If we included a price pressure term, with Π positive definite, the nontrading region would be unchanged, but if we started outside the nontrading region we would trade to another point outside the nontrading region, not to the boundary. We can usually think of this as trading partway to the nontrading region, although not necessarily in the same direction and we can even construct examples in which we would trade further from the nontrading region with

price pressure than without.

3.2.1 Futures overlays

Figure 3 and Figure 4 give additional examples with pure proportional costs, but with parameter values motivated by a futures overlay strategy. Futures overlays are transaction-cost-aware strategies which use futures as an inexpensive way of keeping effective asset allocations in line with a pre-specified allocation. For example, if we think the ideal weighting in equities is 60%, then as the market rises we become over-weighted and as the market falls we become under-weighted (since the fixed-income part of the portfolio moves less than proportionately with moves in the equity market). Maintaining a weighting near the ideal one by trading equities is very expensive. A futures overlay might correct for minor deviations by trading in futures, which are highly correlated with equities but much cheaper to trade.¹⁶ This is why traditional strategies substitute futures trades for trades in equities. However, the expected returns (“alphas”) of these trades are not usually discussed much, but they turn out to be very important. Because most practitioners are not thinking carefully about alphas, they usually follow a symmetric futures overlay, in which futures play a similar role in responding to over-weighting or under-weighting in equities. However, the analysis in this section suggests that in a typical situation, the fund should sell futures to remedy over-weighting, but buy equities to remedy under-weighting.

In our two examples for analyzing futures overlays, we assume realistically that the futures and equities are very highly correlated and that the cost of trading futures is much less than the cost of trading the underlying equities. Consequently, the non-trading region is much smaller in the direction of increasing both than in the direction of increasing one and decreasing the other. Figure 3 illustrates a case in which the investor gets similar expected returns on the underlying and on “synthetic equity” composed of futures plus the riskless asset. This implies that if we are over- or under-weighted in equities initially (with no

¹⁶Perhaps futures are used to keep the exposure to equities within 5% of the target allocation with trading in actual shares of stock only when the allocation gets more than 10% out of line.

futures), we correct our position by selling or buying futures. The strange thing about this example is that there is really no reason to hold equities in the first place. Given that holding futures plus the riskless asset gives the same expected return and almost the same risk as equities, but with a lower cost, a traditional synthetic equity strategy would be optimal, and there would seem to be little reason to hold the underlying equities in the first place. We can see this in Figure 3): if we start from a pure cash position, the point $(0, 0)$ with no equities or futures, it is optimal to follow a “synthetic equity” strategy of buying only futures. And, it is also never optimal to trade equities unless your equities have a weight of less than -15% of your portfolio or more than 100%. The traditional symmetric futures overlay is justified, but only in a case when it is hard to justify holding or trading equity.

Normally, however, we might expect the return on the underlying equities to be higher than the return on synthetic equity due to the benefits of active management. Figure 4 illustrates an example that is the same as the example in Figure 3 except that equities have a significantly higher expected return than the synthetic equity strategy using futures. This assumption does make it optimal to hold equities, and in particular starting with all cash (the point $(0, 0)$) it is optimal to buy only equities. However, in this case the optimal futures overlay is no longer symmetric. The reason for this is that buying equities gives a higher expected return than buying futures, while selling equities give a lower expected return than selling futures. This asymmetry in returns implies an asymmetry in the optimal policy, and for most of the relevant region, it is optimal to sell futures to decrease exposure to the equity market, but it is optimal to buy equities to increase exposure. In the example, once we have a very long position in equities balanced by a very short position in futures, there comes a point when we will sell equities because the long-short combination is not hedged well enough. Hence, the optimal policy will first deal with excess exposure to equities by selling futures, and then will switch to selling equities.

The mean-variance analysis helps us to understand what to expect from a formal analysis of the optimal policy in a continuous-time model. In a continuous-time model, we should think of the cost of rolling one futures contract into the next. Traditionally, practitioners

have thought about the cost of rolling a futures position as a drag on the expected return that is present whether we are holding a long position or a short position,¹⁷ in essence replacing the lumpy cost at roll dates with a constant cost per unit time. While this view would make sense for a position to be held and rolled unconditionally for a long period, it does not make sense for a futures overlay program. If we want to reduce the size of either a long or a short position in futures, it usually makes sense to wait until the roll date, at which time we can unwind the position for free. There is a possibility that the anticipated roll cost would make trading equities relatively more attractive than trading futures, and intuitively this decision depends on the optionality in the purchase, and the fact that the rollover cost could mean that a net position in equities is cheaper for holding a long time but a position in futures is cheaper for holding a short time. Selling futures means that we have an option to reverse the position cheaply, and this is where we should go first if we are over-weighted in equity exposure. However, if we have already bought a lot of futures, the optionality in additional purchase is worth less because we would have to burn through selling our current position before the option is valuable, and we are better off selling equity and not bearing the futures roll cost over time. To first approximation, this strategy (1) only unwinds a futures position, long or short, at the roll date (an exception could occur in a huge market move), (2) instead of buying equities, wait a little while if there is a short futures position that can be exited for free at maturity, (3) otherwise, when justified by the cost of trading, buy equities to increase exposure to the market or sell futures to reduce exposure, subject to a maximum futures exposure as a fraction of the equity exposure.

3.3 Fixed costs

3.3.1 Overall fixed costs

Figure 5 shows a representative example with pure overall fixed costs: $K(S)$ is the same positive cost for all $S \neq \emptyset$, $C_+ = C_- = 0$ (no proportional costs), $\lambda > 0$ (risk averse),

¹⁷See, for example, Brown and Scholtz (2007).

$\kappa = 0$ (no penalty for tracking error), $\Pi = 0$ (no price pressure), and V positive definite (no redundant assets). If there is an overall fixed cost, then if we trade at all we trade to the ideal portfolio (indicated by the blue dot), because the cost is the same whatever trade we make. Consequently, the choice problem reduces to one of comparing the utility of not trading with the utility of trading to the ideal portfolio but incurring the fixed cost. It is optimal to trade when outside the indifference curve (an ellipse or its analogue in some other number of dimensions given our mean-variance assumptions) and the NTR is the yellow area inside of it. If the asset returns were uncorrelated with identical variances, the NTR would be a circle. In this example, everything is symmetric but there is correlation. The positive correlation means that the two securities are substitutes and it is not so bad if we have too little of one asset if we have too much of the other. As in the case of proportional costs, this is why we are quicker to trade if we are over-weighted in both securities than if we are over-weighted in one and under-weighted in the other. If we added quadratic costs, the non-trading region would be larger, and whether we trade to inside, outside, or the boundary of the nontrading region would depend on the starting portfolio. Intuitively, this is because with both fixed and quadratic costs the marginal cost of trading is decreasing initially and increasing afterwards, so we might want to trade further after we reach the boundary or we might have already traded too far.

3.3.2 Security-specific fixed costs

Figure 6 shows a representative example with pure security-specific fixed costs: $K(S) = \sum_{n \in S} K(\{n\})$. The NTR surrounds the ideal portfolio (identified by the blue dot). The upper left and lower right boundaries are on the same ellipse, and the other boundaries are linear. From the regions in the corners, we trade both securities to the ideal portfolio. From the regions on the right and left, we trade Security 1 but not Security 2 to a line going through the ideal portfolio. This would be a vertical line if we had no correlation, but has negative slope in our case because we have positive correlation: since the two risky assets are substitutes, we want to hold more of one if we have less of the other. Similarly, from

the regions above and below, we trade only Security 2 to a different line running through the ideal portfolio. The left plot illustrates the trades in both securities, the middle plot illustrates the trades in Security 1 alone, and the right plot illustrates the trades in Security 2 alone. This example is consistent with Brennan (1975) or Goldstein (1979), in the presence of constraints proportional to the number of securities traded, or implicitly Jacob (1974), where we are given exogenously a small number of securities that can be bought. This analysis is more general mostly in that it does not assume the starting position in cash alone.

3.4 Security-specific fixed and proportional costs

Figure 7 shows a representative example with both security-specific fixed costs and security-specific proportional costs: $K(S) = \sum_{n \in S} K(\{n\})$, $C_+ \gg 0$, $C_- \gg 0$, $\lambda > 0$ (risk averse), $\kappa = 0$ (no penalty for tracking error), $\Pi = 0$ (no price pressure), and V positive definite (no redundant assets). The NTR, resembles the one from Figure 6 when costs are only security-specific fixed and analogous geometrical interpretations apply. The upper left and lower right boundaries are on two different ellipses where we are indifferent between simultaneously trading both securities or staying put, and the other boundaries are straight indifference lines where we are indifferent between trading in one security at a time or doing nothing. Optimal trades for initial allocations outside of the NTR are very similar to those described in Figure 2 when costs are only proportional but end up in the interior of the NTR. This is because the impact of proportional over fixed costs increases with the distance from the ideal portfolio (identified by the blue dot), and when a position is far enough proportional costs are just what matter. In particular, as the leftmost plot illustrates, from the regions in the corners we trade both securities to the closest corner of the NTR defined by the intersections of the dashed lines: the region and the optimal allocations are the same as the one described in Figure 2. While, as shown in the middle and right plot respectively, from regions to the left and right and above and below the NTR, we trade only in security 1 or 2. These trades bring us to the closest dashed straight line inside the NTR, part of which representing the border of the NTR in Figure 2. The reason why the lines stretch beyond is because in these

portions only the position of one security is far enough from the optimum for the fixed cost to be considered sunk, resulting in extra trades involving the most displaced asset.

4 Analytical characterization of the solutions

4.1 Pure proportional costs

When $\kappa + \lambda > 0$, $(\forall S)K(S) = 0$, and $\Pi = 0$, the cost of trading is proportional but perhaps dependent on the direction of trade for each security, and the choice problem is

Problem 2 *Given the vector θ_0 of initial risky security positions, choose a vector of security positions to maximize the utility of terminal wealth*

$$U(\theta) = \theta' \mu - \frac{\lambda}{2} \theta' \mathbf{V} \theta - \frac{\kappa}{2} (\theta - \theta_B)' \mathbf{V} (\theta - \theta_B) - C'_+(\theta - \theta_0)^+ - C'_-(\theta_0 - \theta)^+ \quad (2)$$

4.1.1 First-order conditions

Given θ_0 , it is easy and fast to solve the pure proportional costs version of Problem 1 numerically for the optimal trades. However, we can learn a lot about the economics by studying the first-order conditions, which also allow us to solve small problems exactly by hand as we show in Section 4.1.3. Since the solution of Problem 1 maximizes the concave non-differentiable function $U(\theta)$, so the first-order conditions for an optimum imply¹⁸ $0 \in \nabla U(\theta)$, where $\nabla U(\theta)$ is the subgradient given by $\{\mu - \lambda \mathbf{V} \theta - \kappa \mathbf{V} (\theta - \theta_B) + y | y_n \in [-C_{-,n}, C_{+,n}] (\forall n)\}$.

We can also write the first-order conditions for not wanting to buy or sell any more of security n as

$$(\mu - \lambda \mathbf{V} \theta - \kappa \mathbf{V} (\theta - \theta_B))_n \leq C_{+,n} \quad (3)$$

¹⁸See Rockafellar (1970), Chapter 23, especially Theorem 23.5 with $x^* = 0$. Note that Rockafellar (1970) deals mostly with minimizing convex functions. The inequality defining the subgradient is reversed when the function is concave and we are performing a maximization.

and

$$-(\mu - \lambda \mathbf{V}\theta - \kappa \mathbf{V}(\theta - \theta_B))_n \leq C_{-,n} \quad (4)$$

These two inequalities say that the marginal benefit of buying or selling security n is less than the marginal cost. The solution also needs to satisfy the complementary slackness conditions that (3) (resp. (4)) holds with equality if $(\theta - \theta_0)_n > 0$ (resp < 0). These complementary slackness conditions automatically hold if $\theta = \theta_0$.

4.1.2 Nontrading region

Absent costs ($C_+ = C_- = 0$), Problem 2 features the standard mean-variance ideal allocation θ_I (adjusted for the presence of the benchmark θ_B)

$$\theta_I = \frac{1}{\lambda + \kappa} \mathbf{V}^{-1}(\mu + \kappa \mathbf{V}\theta_B) \quad (5)$$

and it is optimal to trade directly to this portfolio whatever the initial allocation θ_0 . As a result, the NTR is the singleton $\{\theta_I\}$.

With transaction costs, however, there is a non-trivial set of portfolios θ_0 too close to θ_I to justify any trade. This set of portfolios satisfies (3) and (4) and is described by the intersection of half-planes of the form

$$(\mathbf{V}_{.,n})'\theta_0 \geq \frac{1}{\kappa + \lambda}(\kappa(\mathbf{V}_{.,n})'\theta_B + (\mu_n - C_{+,n})) \quad (6)$$

$$(\mathbf{V}_{.,n})'\theta_0 \leq \frac{1}{\kappa + \lambda}(\kappa(\mathbf{V}_{.,n})'\theta_B + (\mu_n + C_{-,n})) \quad (7)$$

where $\mathbf{V}_{.,n}$ is the n -th column of $\mathbf{V}$. These equations with equalities were used to plot the boundaries of the NTRs in Figures 2, 3 and 4. More generally, solving for being on the edge of multiple half-spaces allows us to compute the vertices and faces of the NTR.

In particular, each asset n is associated with half-spaces (6) and (7), which are parallel to each other and have orientation characterized by the vector $\mathbf{V}_{.,n}$. These half-spaces are

separated by the distance $\frac{C_{+,n}+C_{-,n}}{\|\mathbf{V}_i\|(\kappa+\lambda)}$ which is proportional to the magnitude of the costs, and are parallel to the coordinate axes if and only if $\mathbf{V}$ is a diagonal matrix. The resulting object is a parallelogram for the case of 2 risky securities, or its higher dimensional analog for a generic number $N > 2$

The first-order conditions in the case of proportional costs fully characterizes the NTR. Their linearity suggests that Leland made a good assumption when conjecturing the linearity of the NTR boundaries in the heuristic solution of his continuous time model (Leland (2000)). Our analysis is also consistent with the formal solution in continuous time with uncorrelated securities of Liu (2004): as in our model, if risky returns are uncorrelated, the sides of his NTR are parallel to the coordinate axes.

4.1.3 A problem that is easy to solve by hand

The solution and the economics of Problem 2 are actually very easy to derive for the case of two risky securities. Let us take a closer look at the example described in Figure 2, for which we have $\lambda = 1, \kappa = 0, \mu_1 = \mu_2 = 0.02, corr = 0.5, V_{1,1} = V_{2,2} = 0.04$ (implying $\mathbf{V} = \begin{bmatrix} 0.04 & 0.02 \\ 0.02 & 0.04 \end{bmatrix}$) and $C_{+,1} = C_{-,1} = C_{+,2} = C_{-,2} = 0.002$. The first order conditions (6) and (7) inform us that the NTR is the set of portfolios satisfying

$$\begin{aligned} 0.04\theta_1 + 0.02\theta_2 & \begin{cases} \geq 0.018 & \text{for buying security 1: } B1 \\ \leq 0.022 & \text{for selling security 1: } S1 \end{cases} \\ 0.02\theta_1 + 0.04\theta_2 & \begin{cases} \geq 0.018 & \text{for buying security 2: } B2 \\ \leq 0.022 & \text{for selling security 2: } S2. \end{cases} \end{aligned}$$

A portfolio in the no-trading region is at the boundary if at least one inequality holds with equality and it is a corner if two hold with equality. For example, the bottom-right corner solves $S1$ and $B2$, so it is $(\frac{13}{30}, \frac{7}{30})$. Similarly, we can calculate the corners $(\frac{11}{30}, \frac{11}{30})$, $(\frac{7}{30}, \frac{13}{30})$ and $(\frac{3}{10}, \frac{3}{10})$ solving $S1$ and $S2$, $B1$ and $S2$ and $B1$ and $B2$.

Once we have computed the four corners of the NTR, we can write down the entire strategy. In the corner regions, we trade both securities to the corresponding corners. For example, if $\theta_{0,1} > 13/30$ and $\theta_{0,2} < 7/30$, it is optimal to trade to the corner $(13/30, 7/30)$ by selling $\theta_{0,1} - 13/30$ units of security 1 and buying $7/30 - \theta_{0,2}$ units of security 2. In the side and top regions, we buy or sell one security until the corresponding inequality is satisfied with equality. For example, if $\theta_{0,1} > 11/20 - \theta_{0,2}/2$ (so S1 is not satisfied at the starting point) and $7/30 \leq \theta_{0,2} \leq 11/30$ (so B2 and S2 are satisfied if we sell enough of security one so that S1 is just satisfied), then it is optimal to sell $\theta_{0,1} - 11/20 + \theta_{0,2}/2$ units of security 1 without trading security 2. Once we determine the NTR and the optimal strategies in the corners and on the top, bottom, and sides, we know the entire strategy.

4.2 Price pressure (quadratic cost)

Practitioners making large trades are concerned with price pressure: buying tends to move prices up and selling tends to move prices down, and either way larger trades receive a less favorable price.¹⁹ This could be due to information as in Glosten and Milgrom (1985) or Bagehot (1971), since the trade could be due to information about the stock. Or, it could be that there is a temporary shortage of liquidity in the market and the price pressure could reflect risk aversion or inventory costs of the market makers. We can think of the impact of price pressure on cash flow as analogous to a trading cost (or as a trading cost if we want to take a broad view). If the increase in price for each stock is proportional to the amount traded (as in the equilibrium in Kyle (1985) and many papers in the subsequent literature), this introduces a quadratic term in the objective function since the cost of buying the change in position is the price times quantity, where the quantity is linear in the price. A quadratic term fits nicely within the mean-variance analysis. Hence, to analyze price pressure we focus on the quadratic cost term $-(\theta - \theta_0)' \Pi (\theta - \theta_0)/2$ in the utility function (1).

¹⁹As usual, matters are somewhat more complicated in practice, and in some markets a trade of intermediate size receives a more favorable price than a small trade, perhaps due to price discrimination against retail customers.

If costs are purely quadratic, the setup is a straightforward extension of the standard mean-variance analysis. Problem 1 specializes to a strictly concave maximization problem admitting a unique solution determined by the first order condition

$$\mu + \kappa \mathbf{V} \theta_B - (\lambda + \kappa) \mathbf{V} \theta - \mathbf{\Pi}(\theta - \theta_0) = 0 \quad (8)$$

From (8) follows that the no-trading region is a singleton containing the ideal portfolio θ_I defined in (5). For any other initial allocation θ_0 the optimal strategy is to trade to

$$\theta_{PI} = [(\lambda + \kappa) \mathbf{V} + \mathbf{\Pi}]^{-1} (\mu + \kappa \mathbf{V} \theta_B + \mathbf{\Pi} \theta_0) \quad (9)$$

Having an optimum that is a single point simplifies analysis in continuous time as in the equilibrium models of Back (1993) and Kyle et al. (2018). Whenever proportional or fixed costs are added, the nontrading region becomes nontrivial.

When there are no fixed costs, there is an interesting connection between the problems with and without price pressure. With just proportional costs and price pressure, the choice problem is

Problem 3 (*Proportional Costs with Price Pressure*) *Given the vector θ_0 of initial risky security positions, choose a vector of security positions to maximize the utility of terminal wealth*

$$\theta' \mu - \frac{\lambda}{2} \theta' \mathbf{V} \theta - \frac{\kappa}{2} (\theta - \theta_B)' \mathbf{V} (\theta - \theta_B) - C'_+(\theta - \theta_0)^+ - C'_-(\theta_0 - \theta)^+ - \frac{1}{2} (\theta - \theta_0)' \mathbf{\Pi} (\theta - \theta_0) \quad (10)$$

For this problem, the first-order conditions (3) and (4) become

$$(\mu - \lambda \mathbf{V} \theta - \kappa \mathbf{V} (\theta - \theta_B) - \mathbf{\Pi} (\theta - \theta_0))_n \leq C_{+,n} \quad (11)$$

and

$$-(\mu - \lambda \mathbf{V} \theta - \kappa \mathbf{V} (\theta - \theta_B) - \mathbf{\Pi} (\theta - \theta_0))_n \leq C_{-,n}. \quad (12)$$

These two inequalities say that the marginal benefit of buying or selling security n is less than the marginal cost. The solution also needs to satisfy the complementary slackness conditions that (11) (resp. (12)) holds with equality if $(\theta - \theta_0)_n > 0$ (resp < 0). The following theorem compares and contrasts the solution with and without price pressure.

Theorem 2 *Consider Problem 3 with price pressure for all trades (Π positive definite) and Problem 2 without price pressure, but with all the same parameters except Π . Then the NTR is the same in both problems. However, if we start outside the NTR, we never trade to a point in the NTR (including its boundary) in Problem 3 with price pressure, but we always trade to a point in the NTR in Problem 2 without price pressure. In particular, this implies that if we trade at all, the optimal trade with price pressure is never the same as the optimal trade without price pressure.*

Proof. When $\theta = \theta_0$, the first-order conditions (3) and (4) are exactly the same as the first-order conditions (11) and (12), since $\theta = \theta_0$ implies $-\Pi(\theta - \theta_0) = 0$. Also, complementary slackness holds in both cases. Since the objective function is strictly concave and the constraint set is convex, these first-order conditions are necessary and sufficient, so both problems have the same solution. This shows the two problems have the same NTR. To show that the optimal solution for Problem 2 is always in the NTR, note that the solution to the problems always satisfies the conditions (3) and (4) that characterize the NTR (and additional complementary slackness conditions). Now we need to show that if θ_0 is not in the NTR, neither is the optimal solution θ to the Problem 3. We argue by contradiction: assume not, and then for some parameter values, the optimal θ is in the NTR but θ_0 is not. This implies θ satisfies the first-order conditions (3) and (4) and the complementary slackness conditions that if $(\theta - \theta_0)_n > 0$ then (3) holds with equality, while if $(\theta - \theta_0)_n < 0$

then (4) holds with equality. Now, let

$$C_n^* \equiv \begin{cases} C_{+,n} & \text{if } (\theta - \theta_0)_n > 0 \\ -C_{-,n} & \text{if } (\theta - \theta_0)_n < 0 \\ 0 & \text{otherwise} \end{cases} \quad (13)$$

Then for all n ,

$$(\theta - \theta_0)_n (\mu - \lambda \mathbf{V}\theta - \kappa \mathbf{V}(\theta - \theta_B) - \Pi(\theta - \theta_0))_n = (\theta - \theta_0)_n C_n^*. \quad (14)$$

When $(\theta - \theta_0)_n > 0$, this follows from complementary slackness of (11); when $(\theta - \theta_0)_n < 0$, this follows from complementary slackness of (12).²⁰ Otherwise, this follows because $(\theta - \theta_0)_n = 0$. Similarly, since we are given that θ is in the NTR, it must be that θ satisfies (3) and (4), which implies that

$$(\theta - \theta_0)_n C_n^* \geq (\theta - \theta_0)_n (\mu - \lambda \mathbf{V}\theta - \kappa \mathbf{V}(\theta - \theta_B))_n, \quad (15)$$

or, summing over n ,

$$(\theta - \theta_0)' C^* \geq (\theta - \theta_0)' (\mu - \lambda \mathbf{V}\theta - \kappa \mathbf{V}(\theta - \theta_B)). \quad (16)$$

However, summing (14) over n implies

$$\begin{aligned} (\theta - \theta_0)' C^* &= (\theta - \theta_0)' (\mu - \lambda \mathbf{V}\theta - \kappa \mathbf{V}(\theta - \theta_B) - \Pi(\theta - \theta_0)) \\ &= (\theta - \theta_0)' (\mu - \lambda \mathbf{V}\theta - \kappa \mathbf{V}(\theta - \theta_B)) - (\theta - \theta_0)' \Pi(\theta - \theta_0) \\ &< (\theta - \theta_0)' (\mu - \lambda \mathbf{V}\theta - \kappa \mathbf{V}(\theta - \theta_B)), \end{aligned} \quad (17)$$

since $\theta \neq \theta_0$ and positive definiteness of Π imply that $(\theta - \theta_0)' \Pi(\theta - \theta_0) > 0$. However, (17) is inconsistent with (16), contradicting our assumption that θ_0 is outside the NTR but θ is

²⁰When $(\theta - \theta_0)_n < 0$, the inequality in (12) is reversed by multiplication by a negative number.

in it. We have shown that if θ_0 is not in the NTR, then the optimal solution to Problem 2 is always in the NTR (on its boundary as shown in Subsection 4.1) but the solution to Problem 3 is never in the NTR. This shows the two solutions cannot be the same, and we are done. ■

Intuitively, in the direction we are trading (including both purchases and sales), we must be indifferent at the margin because we are free to go a little further or a little less. However, with price pressure the marginal cost is increasing in how much we trade, if we are indifferent at the margin making our trade, we must strictly prefer to trade at the margin if we start there. If we assumed Π to be positive semi-definite rather than positive-definite, an optimal trade in Problem 3 that produces price pressure $((\theta - \theta_0)' \Pi (\theta - \theta_0) > 0)$ cannot trade to the nontrading region, with the same proof as for Theorem 2, except that the inequality (17) follows from the assumption $(\theta - \theta_0)' \Pi (\theta - \theta_0) > 0$.

4.3 The utility loss from transaction costs

Constantinides (1986) makes an interesting argument that the impact of proportional transaction costs is second-order because agents tend to react to an increase in transaction costs by trading less. That analysis was in a continuous-time model with a stock price following a standard lognormal diffusion. Dai et al. (2016) show a contrary result that the impact is first-order if there are predictable trading stops, e.g. over a weekend, which is formally the same as allowing jumps in prices at predictable times.²¹ Jang et al. (2007) show a similar result that if the return parameters follow a Markov switching model, the impact of costs is first-order. In our model, we have an exact expression for the utility loss as a function of cost and this helps to understand these results and more generally the impact on the agent of different types of trading costs. We measure the impact by utility loss compared with the

²¹To make the change in price over the weekend looks like a jump at a point in time, change the time index to delete the time during the trading halt and give the times at the beginning and end of the trading halt the same time index. To make this rigorous, we could make the price process right-continuous and left-limiting and the portfolio processes left-continuous and right-limiting, as in Dybvig and Huang (1988).

optimum in a model with no costs.²² Our analysis shows that unless we start at the ideal point, the impact of cost is first-order for small costs and constant for large costs. Small costs do not change the optimal trade by very much, and given the trade, the utility loss is approximately proportional to the size of the cost. For large costs, it does not pay to trade much, and the utility loss is approximately equal to the utility loss of not trading. These results hold in our single period model whether costs are proportional, quadratic, or fixed, or a combination of the three. The results should hold in various continuous-time settings as well: if costs are small and we do not start at the ideal point, we are outside the nontrading region for small enough cost, and the utility loss from trading to the nontrading region is first order because, given small costs, it is approximately equal to the utility loss from trading to the ideal point. Similarly, if costs are very large, the utility loss is approximately constant because it is almost the same as the utility loss of not trading at all, provided that is feasible and all trading incurs some cost.

If we start at the ideal point, the utility loss in our single-period model is zero whatever the cost, but this result is special to the single-period model. A lot of the trading in practice is rebalancing because price movements move you away from what would be optimal absent costs. For example, in an iid lognormal world, an agent with constant relative risk aversion would hold a fixed proportion in stocks and bonds absent costs, but stocks and bonds have different sensitivities to market moves, which implies that the proportions change when the market moves. Starting at the ideal portfolio is the case in which the Constantinides (1986) result is probably exactly true: there is no cost in the single-period model but given the lognormal diffusion the cost is second-order. The first-order impact of costs found by Dai et al. (2016) and Jang et al. (2007) still hold even if we start at the ideal portfolio, because price movements during the market closure or lumpy new information arrival will sometimes move the portfolio outside the nontrading region, and the cost of trading back to the nontrading region is first-order.

²²The other papers focus on the impact on returns, which has the same units as the utility loss for our mean-variance preferences.

Now we turn to a formal analysis of the utility loss as a function of the scale of costs. We define the scale of costs to be a multiplier γ of all the costs, so that the objective function now becomes

$$U(\theta, \gamma) = \theta' \mu - \frac{\lambda}{2} \theta' \mathbf{V} \theta - \frac{\kappa}{2} (\theta - \theta_B)' \mathbf{V} (\theta - \theta_B) - \gamma (C'_+(\theta - \theta_0)^+ + C'_-(\theta_0 - \theta)^+ + K(\{n | (\theta - \theta_0)_n \neq 0\})) + \frac{1}{2} (\theta - \theta_0)' \mathbf{\Pi} (\theta - \theta_0). \quad (18)$$

so we can think of $\gamma = 1$ in Problem 1. We are mostly concerned about the behavior of the utility loss as a function of γ compared to the no-cost problem ($\gamma = 0$) in which it is optimal to trade to the ideal portfolio θ_I defined in (5). In this subsection, we assume that $\mathbf{V}$ is positive definite so that $\mathbf{V} + \gamma \mathbf{\Pi}$ will be positive definite for all $\gamma \geq 0$.

In the next two theorems, we prove that the utility loss compared to zero transaction costs is roughly proportional to the scale of costs when the costs are close to zero and roughly constant when costs are high. This is because when costs are low, it is approximately optimal to trade to the ideal point that would be optimal absent costs, and when costs are high, it is approximately optimal not to trade.

Theorem 3 *Suppose that $\mathbf{V}$ is positive definite and that $K(S)$ is nondecreasing as we add securities to S . Then, as γ approaches zero, the portfolio that maximizes $U(\theta, \gamma)$ converges to θ_I , and to first order in γ the utility loss is equal to the utility loss $\gamma(C'_+(\theta_I - \theta_0)^+ + C'_-(\theta_0 - \theta_I)^+ + K(\{n | (\theta_I - \theta_0)_n \neq 0\})) + (\theta_I - \theta_0)' \mathbf{\Pi} (\theta_I - \theta_0)/2$ from trading to θ_I .*

Proof. We will derive the first-order utility loss by bounding the utility loss above and below by functions that are the same to the first order in γ . For the lower bound on the utility loss (from an upper bound on utility), we modify the cost function so that there is no additional fixed or marginal cost for trading securities not already traded by going from θ_0 to θ_I . This relaxed cost function makes it so we don't have to worry about complementarity slackness conditions or incremental fixed cost in any securities for which $\theta_I - \theta_0 = 0$, but the optimum still gives a tight enough bound to pin the rate of utility loss as we increase γ .

Define C^* by

$$C_n^* \equiv \begin{cases} C_{+,n} & \text{if } (\theta_I - \theta_0)_n > 0 \\ -C_{-,n} & \text{if } (\theta_I - \theta_0)_n < 0 \\ 0 & \text{otherwise} \end{cases} \quad (19)$$

Let $S_I \equiv \{n | (\theta_I - \theta_0)_n \neq 0\}$ be the set of securities that are traded to move from θ_0 to θ_I . Then, we have the modified cost function for any portfolio θ for which $(\theta - \theta_0)_n$ has the same sign as $(\theta_I - \theta_0)_n$ whenever $(\theta_I - \theta_0)_n \neq 0$:

$$\gamma(C^{*'}(\theta - \theta_0) + K(S_I) + \frac{1}{2}(\theta - \theta_0)' \mathbf{\Pi}(\theta - \theta_0)), \quad (20)$$

and we can write the objective function as

$$U(\theta, \gamma) = U(\theta_I, 0) - \frac{\kappa + \lambda}{2}(\theta - \theta_I)' \mathbf{V}(\theta - \theta_I) - \gamma(C^{*'}(\theta - \theta_0) + K(S_I) + \frac{1}{2}(\theta - \theta_0)' \mathbf{\Pi}(\theta - \theta_0)), \quad (21)$$

where $U(\theta_I, 0)$ is the utility of the ideal portfolio absent costs (with $\gamma = 0$). Assuming trade is in the same direction for nonzero trades to θ_I is correct for us, because that will be true when γ is small enough, as we show shortly. The first-order condition for an optimum in this case is

$$-(\kappa + \lambda)\mathbf{V}(\theta^* - \theta_I) - \gamma C^* - \gamma \mathbf{\Pi}(\theta^* - \theta_0) = 0, \quad (22)$$

and the optimal portfolio is therefore

$$\theta^* = \theta_I - \gamma((\kappa + \lambda)\mathbf{V} + \gamma \mathbf{\Pi})^{-1}(C^* + \mathbf{\Pi}(\theta_I - \theta_0)). \quad (23)$$

We can thus see that θ^* converges to θ_I as γ goes to 0, justifying our assumption that $\theta^* - \theta_0$ has the same sign as nonzero elements of $\theta_I - \theta_0$ when γ is small enough. The utility from

this optimal portfolio can now be written as

$$\begin{aligned}
U(\theta^*, \gamma) &= U(\theta_I, 0) - \gamma(C^{*'}(\theta_I - \theta_0) + K(S_I) + \frac{1}{2}(\theta_I - \theta_0)' \mathbf{\Pi}(\theta_I - \theta_0)) \\
&\quad + \frac{1}{2}(\theta^* - \theta_I)'((\kappa + \lambda)\mathbf{V} + \gamma\mathbf{\Pi})(\theta^* - \theta_I).
\end{aligned} \tag{24}$$

The first term is the utility of trading to θ_I when there are no costs, the second the cost of trading to θ_I , and the third term is the quadratic form saying how much better off you are at θ^* than at θ_I , given costs. (Note that because $(\kappa + \lambda)\mathbf{V} + \gamma\mathbf{\Pi}$ is positive-definite, $(\theta^* - \theta_I)'((\kappa + \lambda)\mathbf{V} + \gamma\mathbf{\Pi})(\theta^* - \theta_I) > 0$). Since $\theta^* - \theta_I$ is $O(\gamma)$, the final term is $O(\gamma^2)$ and we conclude that $dU(\theta^*, \gamma)/d\gamma|_{\gamma=0} = -C^{*'}(\theta - \theta_0) + K(S_I) + \frac{1}{2}(\theta - \theta_0)' \mathbf{\Pi}(\theta - \theta_0)$. Thus, for γ small, the relaxed utility (24) is an upper bound for the optimal value of (18) due to the lower cost. However, for γ small, the first two terms (without the third) give a lower bound for the optimal value of (18), since they give the value of trading to θ_I (without the additional gain from trading to θ^* instead). Since this lower bound is the same to first order in γ as the upper bound, so is the the optimal value of (18). Therefore, for small γ , the utility loss is approximately proportional to the scale γ of costs, and equal at first order to the cost of trading to the ideal portfolio. The constant of proportionality is zero only when the cost of trading to the ideal portfolio θ_I is zero. ■

Theorem 4 *Suppose that $\mathbf{V}$ is positive definite, that $K(S)$ is nondecreasing as we add securities to S , and that all (nonzero) trading is costly. Then, as γ increases without bound, the portfolio converges to θ_0 (no trading) and the utility loss converges to the utility loss $U(\theta_I, 0) - U(\theta_0, 0)$ from not trading.*

Proof. For a large cost, we need to show that as γ increases, the utility loss converges to a constant equal to the utility loss of the best trade that can be made at no cost. First, we show that for γ large enough, we will not incur any fixed cost. If $K(S) \equiv 0$, this is obvious, since there are no fixed costs. Otherwise, set $\bar{\gamma} = (U(\theta_I) - U(\theta_0))/\min\{K(S)|K(S) > 0\}$. Note that the minimum exists and is positive because there are only finitely many subsets

of the set of securities. Then, we will show that for all $\gamma > \bar{\gamma}$, any portfolio θ that bears a positive fixed cost is dominated by not trading:

$$\begin{aligned}
U(\theta, \gamma) &\leq U(\theta_I, 0) - \gamma K(\{n | (\theta - \theta_0)_n \neq 0\}) \\
&< U(\theta_I, 0) - \bar{\gamma} \min\{K(S) | K(S) > 0\} \\
&= U(\theta_I, 0) - (U(\theta_I, 0) - U(\theta_0, 0)) \\
&= U(\theta_0, 0).
\end{aligned}$$

Consider a sequence $\gamma_k \rightarrow \infty$ and corresponding optimal portfolios θ_k that maximize (18). If some subsets bear fixed costs (so $K(S)$ is not identically 0), we can restrict attention to the later part of the sequence when $\gamma_k > \bar{\gamma}$, so we can assume without loss of generality that θ_k 's do not bear any fixed costs. We can write (18) as

$$U(\theta_k, \gamma) = U(\theta_k, 0) - \gamma T(\theta_k) - \gamma K(\{n | (\theta_k - \theta_0)_n \neq 0\}). \quad (25)$$

$U(\theta_k, 0)$ is the utility absent costs, which is a strictly concave quadratic function with a maximum at θ_I . $T(\theta_k)$ includes proportional and quadratic costs, and is nonnegative, convex, continuous, and piecewise quadratic. The final fixed-cost term is zero for each θ_k , and we will show it is also zero in the limit. All the θ_k 's lie in the set $\Theta \equiv \{\theta | U(\theta, 0) \geq U(\theta_0, 0)\}$ (which is nonempty and compact by Lemma 1), since portfolios outside this set are dominated by not trading and picking $\theta = \theta_0$, which does not incur any cost.

Since all the θ_k 's are in the compact set Θ , there exists a convergent subsequence $k_j \rightarrow \infty$ and a limit $\hat{\theta}$ such that $\theta_{k_j} \rightarrow \hat{\theta}$. On any convergence subsequence, $U(\theta_{k_j}, 0)$ and $T(\theta_{k_j})$ both converge (because the functions are continuous). Also, $T(\theta_{k_j}) \rightarrow 0$, since otherwise $U(\theta_{k_j}, \gamma) = U(\theta_{k_j}, 0) - \gamma T(\theta_{k_j}) \rightarrow -\infty$, which would contradict optimality of θ_{k_j} for sufficiently large j , for which choosing θ_{k_j} would be dominated by choosing θ_0 since $U(\theta_0, \gamma_{k_j}) = U(\theta_0, 0)$ is finite. Also, $K(\{n | (\hat{\theta} - \theta_0)_n \neq 0\}) = 0$ since we restricted attention to portfolios for which the fixed cost is zero ($K(\{n | (\theta_{k_j} - \theta_0)_n \neq 0\}) = 0$) and $K(S)$ is

nonnegative and nondecreasing in S .²³ Therefore, every limit point $\hat{\theta}$ of θ_k has zero cost: $T(\hat{\theta}) + K(\{n | (\hat{\theta} - \theta_0)_n = 0\}) = 0$. And, by assumption, $\theta = \theta_0$ (not trading) is the only portfolio for which there is no trading cost. Furthermore, θ_k must converge to θ_0 , not just on a subsequence, because otherwise there is a subsequence that stays some distance away from θ_0 , and this sequence will have a convergent subsequence that converges to something other than θ_0 , which is a contradiction to our result that all convergent subsequences converge to θ_0 .

Since θ_k converges to θ_0 and $U(\theta, 0)$ is continuous, we have that $\lim_{k \uparrow \infty} U(\theta_k, \gamma_k) = \lim_{k \uparrow \infty} (U(\theta_k, 0) - \gamma T(\theta_k)) \leq U(\theta_0, 0)$. The inequality is an equality, since otherwise, not trading ($\theta = \theta_0$) would dominate θ_k for all sufficiently large k , contradicting optimality of θ_k for γ_k . ■

5 Conclusion

Many frictions in financial markets, for example, the bid-ask spread, brokerage fees, price pressure, or the cost of attention, can be modeled by assuming that agents face transaction costs with various functional forms. We have studied transaction costs in a single-period mean-variance setting. This setting allows a full solution for many securities with possibly correlated returns, and makes the economics clear. The impact on trading costs on the optimal solution is qualitatively quite different depending on whether the marginal cost of trading in some direction is decreasing, constant, or increasing. We represent these three cases by considering fixed, proportional, and quadratic trading costs, and combinations of the three. Except in the case of pure quadratic costs (which might be due to price pressure), trading costs in our model imply the existence of a nontrivial nontrading region (NTR) of portfolios from which no trade is justified.

For fixed costs with or without proportional costs, the marginal cost of trading is de-

²³Because the sequence θ_{k_j} converges to $\hat{\theta}$, then for j sufficiently large, $\theta_{k_j} - \theta_0$ has the same sign as $\hat{\theta} - \theta_0$ for all nonzero elements of $\hat{\theta}$. Therefore, for j large enough, $0 \leq K(\{n | (\hat{\theta} - \theta_0)_n \neq 0\}) \leq K(\{n | (\theta_{k_j} - \theta_0)_n \neq 0\}) = 0$

creasing and any trade goes to inside the NTR, while for quadratic costs with or without proportional costs, the marginal cost of trading is increasing and any trade goes to outside the NTR. If there are only proportional costs any trading is to the boundary of the NTR. With overall fixed costs plus proportional costs, the trade is the same as it would be without the fixed costs if we start far from the ideal portfolio we would hold absent any cost, since given that we will cover the fixed cost we can view it as sunk.²⁴ If we have quadratic costs plus proportional costs, the NTR is the same as it would be without the quadratic costs, since the marginal quadratic cost is zero when the net trade is zero.

The analysis provides new insights into practical problems such as the use of futures overlays. Futures overlays do some portfolio rebalancing in futures instead of the underlying stocks as a way of reducing transaction costs. Our analysis shows that the optimal strategy should typically be asymmetric, buying stocks to increase exposure and shorting futures to reduce exposure. This is in contrast to traditional symmetric overlay strategies.

Our results can be used to understand the claim by Constantinides (1986) that the impact of transaction costs is second-order because to first order agents react by trading less and counterclaims (Dai et al. (2016) and Jang et al. (2007)) that the impact is first-order. In our model, unless we start at the ideal portfolio, the utility loss from transaction costs is roughly proportional to cost when costs are small and roughly constant when costs are large. This is because if costs are small our trade is approximately to the ideal portfolio and increasing costs increase utility loss approximately proportionally, while if costs are large, we have approximately no trade, and the loss is insensitive to cost. If we start at the ideal portfolio, the cost is zero in our single-period model but typically not in a multiperiod model because in some realizations it will be optimal to trade later. This is the case in which the second-order impact of Constantinides (1986) could happen. However, as pointed out by the other papers, if we add some uncertainty (such as jumps in the price process, price moves during a market closure, or jumps in a state variable driving returns) that could land the portfolio

²⁴This result is due in part to the fact that the mean-variance preferences we assume, like CARA preferences, do not have a wealth effect.

outside the non trading region, the cost will again be proportional at low cost because of the cost of moving back near the ideal portfolio. In general, the economic argument coming from our model suggests that the usual case has proportional impact at low costs and constant impact at high costs, and that this is true not only for proportional costs but also for any combination of fixed, proportional, and quadratic costs.

A variant of our analysis has been used by Maurer et al. (2020) to look at transaction-cost smart futures strategies. The strategy improves a lot on the optimal strategy ignoring costs and more interestingly the optimal strategy ignoring correlations among futures. This shows that our extension to multiple securities that can be correlated is important economically. Another attractive empirical feature of our setup is the ease at which it scales up with the number of securities: Pezzo et al. (2020) construct profitable out-of-sample trading strategies in the U.S. stock market involving thousands of stocks.

Another possible application of this analysis is in examining whether asset pricing “anomalies” can be used to generate profits. In a world with trading costs, anomalies can exist even if all agents are optimizing and rational, provided the profitability of trading does not cover the costs. With this analysis, we can shine a fresh light on whether trading on anomalies is profitable. Possibly the smarter trading in the face of transaction costs will reveal that trading on anomalies is more profitable than previously thought, or perhaps we will find out that trading on anomalies gets unprofitable exactly when naive analysis (say using a linear forecasting model) suggests that profits are the largest.

Although the analysis is fairly general, further generality is possible even without going outside the tractable quadratic programming framework in the paper. As in Markowitz (1959), we can impose various linear constraints, including many constraints that are likely to be imposed by or on portfolio managers such as nonnegative holdings in all securities, no borrowing, no more than twice the index weight in any industry, no more than 5% of the portfolio in one company’s stock, etc. It is also possible to impose a condition that the cost of establishing a short position is different from the cost of selling out of holdings, as in an early draft of Maurer et al. (2020). In their analysis, an agent can sell part or all of a

long position with one marginal cost, but any selling beyond that incurs the larger marginal cost associated with establishing a short position. More generally, we can write down an activity model to incorporate various sorts of security swaps and bundles trading as different activities (in additions to the activities of buying or selling individual securities).

The analysis provides insights into continuous-time problems that are hard to solve exactly. For example, the high-dimensional parallelogram shape of the nontrading region assumed in the heuristic analysis by Leland (2000) is exactly correct in our model, lending support to this assumption. We hope that insights from our analysis will help scholars to push the continuous-time model further.

6 Appendix

Here is a basic result about quadratic forms that is used for proving existence of a solution of the rebalancing problem.

Lemma 1 *Let a be a scalar, let b be an $n \times 1$ vector, let A be a symmetric negative-definite matrix, and let θ_0 be a fixed vector in $\mathbb{R}^n$. Then $\mathcal{S} = \{\theta \in \mathbb{R}^n | a + \theta'b + \theta'A\theta \geq a + \theta_0'b + \theta_0'A\theta_0\}$ is a compact nonempty set.*

Proof. $\mathcal{S}$ is nonempty because $\theta_0 \in \mathcal{S}$. In $\mathbb{R}^n$, a set is compact if and only it is closed and bounded. The set $\mathcal{S}$ is closed because it is the inverse image of the closed set $\{x | x \geq a + \theta_0'b + \theta_0'A\theta_0\}$ under the continuous function $f(\theta) \equiv a + \theta'b + \theta'A\theta$. It only remains to show that $\mathcal{S}$ is bounded. Since A is symmetric and negative definite, all of A 's eigenvalues are real and negative. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of A with corresponding orthonormalized eigenvectors $x_1, x_2, \dots, x_n$, and let $\theta^* \equiv -A^{-1}b$ be the vector that maximizes $f(\theta)$. Furthermore, let X be the $n \times n$ matrix whose rows are the normalized eigenvectors and let D be the diagonal matrix whose diagonal elements are the corresponding eigenvalues, so that $A = X'DX$. Then, since $f(\theta) = f(\theta^*) + (\theta - \theta^*)'A(\theta - \theta^*)$, every element $\theta \in \mathcal{S}$ satisfies

$f(\theta_0) \leq f(\theta)$ and therefore

$$f(\theta_0) - f(\theta^*) \leq f(\theta) - f(\theta^*) \quad (26)$$

$$= (\theta - \theta^*)' A (\theta - \theta^*) \quad (27)$$

$$= (\theta - \theta^*)' X' D X (\theta - \theta^*) \quad (28)$$

$$= \sum_i \lambda_i ((\theta - \theta^*)' x_i)^2 \quad (29)$$

$$\leq \max\{\lambda_i\} \sum_i ((\theta - \theta^*)' x_i)^2 \quad (30)$$

$$= \max\{\lambda_i\} (\theta^* - \theta)' X' I X (\theta^* - \theta) \quad (31)$$

$$= \max\{\lambda_i\} (\theta^* - \theta)' (\theta^* - \theta), \quad (32)$$

where the last step used the fact that $X'X = I$ since we chose orthonormal eigenvectors. Therefore, noting the maximum of the λ_i 's is negative, $\|\theta^* - \theta\|^2 = (\theta^* - \theta)'(\theta^* - \theta) \leq (f(\theta^*) - f(\theta_0))/(-\max\{\lambda_i\})$, and consequently

$$\|\theta\|^2 \leq \|\theta^*\|^2 + \|\theta^* - \theta\|^2 \quad (33)$$

$$\leq \|\theta^*\|^2 + (f(\theta^*) - f(\theta_0))/(-\max\{\lambda_i\}), \quad (34)$$

which completes the proof that $\mathcal{S}$ is bounded. ■

References

- Back, Kerry, 1993, Asymmetric Information and Options, *The Review of Financial Studies* 6, 435–472.
- Bagehot, Walter, 1971, The only game in town, *Financial Analysts Journal* 22, 12–14.
- Balduzzi, Pierluigi, and Anthony W. Lynch, 1999, Transaction costs and predictability: some utility cost calculations, *Journal of Financial Economics* 52, 47–78.
- Balduzzi, Pierluigi, and Anthony W. Lynch, 2000, Predictability and Transaction Costs: The Impact on Rebalancing Rules and Behavior, *Journal of Finance* 55, 2285–2309.
- Bawa, Vijay S., Stephen J. Brown, and Roger W. Klein, 1979, Estimation Risk and Optimal Portfolio Choice, in *Estimation Risk and Optimal Portfolio Choice* (North-Holland, New York).

- Brennan, M., 1975, The Optimal Number of Securities in a Risky Asset Portfolio When There are Fixed Costs of Transacting: Theory and Some Empirical Results, *Journal of Financial and Quantitative Analysis* 10, 483–496.
- Brown, T. David, and Daniel Scholtz, 2007, Rebalancing Revisited: The Role of Derivatives, *Financial Analysts Journal* 63, 32–44.
- Buss, Adrian, and Bernard Dumas, 2017, The Dynamic Properties of Financial-Market Equilibrium with Trading Fees, Working paper, INSEAD.
- Constantinides, George M., 1986, Capital market equilibrium with transaction costs, *Journal of Political Economy* 94, 842–842.
- Dai, Min, Peifan Li, Hong Liu, and Yajun Wang, 2016, Portfolio Choice with Market Closure and Implications for Liquidity Premia, *Management Science* 62, 368–386.
- Davis, Mark H.A., and A. R. Norman, 1990, Portfolio Selection With Transaction Costs, *Mathematics of Operations Research* 15, 676–713.
- Donohue, C., and K. Yip, 2003, Optimal Portfolio Rebalancing with Transaction Costs: Improving on Calendar- or Volatility-Based Strategies, *Journal of Portfolio Management* 29, 49–63.
- Dumas, Bernard, and Elisa Luciano, 1991, An Exact Solution to a Dynamic Portfolio Choice Problem under Transactions Costs, *Journal of Finance* 46, 577–595.
- Dybvig, Philip H., and Chi-fu Huang, 1988, Nonnegative Wealth, Absence of Arbitrage, and Feasible Consumption Streams, *Review of Financial Studies* 1, 377–401.
- Garleanu, Nicolae, and Lasse H. Pedersen, 2013, Dynamic Trading with Predictable Returns and Transaction Costs, *Journal of Finance* .
- Glosten, L.R., and P.R. Milgrom, 1985, Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders, *Journal of Financial Economics* 14, 71–100.
- Goldstein, A., 1979, A Model of Capital Asset Pricing with Transaction Costs, Ph.d dissertation, Yale University.
- Goodman, Jonathan, and Daniel N. Ostrov, 2010, Balancing Small Transaction Costs with Loss of Optimal Allocation in Dynamic Stock Trading Strategies, *SIAM Journal on Applied Mathematics* 70, 1977–1998.
- Grinold, Richard C., and Ronald N. Kahn, 1995, Active Portfolio Management, Technical report, Probus: Chicago.
- Gu, Shihao, Bryan Kelly, and Dacheng Xiu, 2020, Empirical Asset Pricing via Machine Learning, *The Review of Financial Studies*, forthcoming .

- Han, Jiarui, 2005, Dynamic Portfolio Management: An Approximate Linear Programming Approach, Ph.d dissertation, Stanford University.
- Harvey, Campbell R., and Goufu Zhou, 1990, Bayesian Inference in Asset Pricing Tests, *Journal of Financial Economics* 26, 221–254.
- Irle, A., and C. Prelle, 2008, A renewal theoretic result in portfolio theory under transaction costs with multiple risky assets, Working paper, Kiel Institute for the World Economy.
- Jacob, N. L., 1974, A Limited-Diversification Portfolio Selection model for the Small Investor, *Journal of Finance* 29, 874–856.
- Jang, Bong-Gyu, Hyeng Keun Koo, Hong Liu, and Mark Loewenstein, 2007, Liquidity Premium and Transaction Costs, *Journal of Finance* 62, 2329–2366.
- Kyle, Albert S., 1985, Continuous Auctions and Insider Trading, *Econometrica* 6, 1315–1335.
- Kyle, Albert S., Anna A. Obizhaeva, and Yajun Wang, 2018, Smooth Trading with Overconfidence and Market Power, *Review of Economic Studies* 85, 611–62.
- Leland, Hayne E., 1985, Option Pricing and Replication with Transactions Costs, *The Journal of Finance* 40, 1283–1301.
- Leland, Hayne E., 2000, Optimal portfolio implementation with transaction costs and capital gain taxes, Working paper, University of California at Berkeley.
- Liu, Hong, 2004, Optimal Consumption and Investment with Transaction Costs and Multiple Risky Assets, *Journal of Finance* 59, 289–338.
- Liu, Hong, and M. Loewenstein, 2004, Optimal Portfolio Selection with Transaction Costs and Finite Horizons, *Review of Financial Studies* 15, 805–835.
- Lynch, Anthony W., and Sinan Tan, 2009, Multiple Risky Assets, Transaction Costs, and Return Predictability: Allocation Rules and Implications for the U.S. Investors, *Journal of Financial and Quantitative Analysis* 45, 1015–1053.
- Mao, J. C. T., 1970, Essentials of Portfolio Diversification Strategy, *Journal of Finance* 25, 1109–1121.
- Markowitz, Harry, 1952, Portfolio Selection, *Journal of Finance* 7, 77–91.
- Markowitz, Harry, 1959, Portfolio selection: efficient diversification of investment, Technical report, New York: Wiley.
- Masters, S. J., 2003, Rebalancing: Establishing a Consistent Framework, *Journal of Portfolio Management* 29, 52–57.
- Maurer, Thomas A., Luca Pezzo, and Mark P. Taylor, 2020, Importance of Transaction Costs for Asset Allocations in FX Markets, Working paper, Washington University.

- Mayshar, J., 1981, Transaction Costs and the Pricing of Assets, *Journal of Finance* 36, 583–597.
- Muthuraman, K., and S. Kumar, 2006, Multidimensional Portfolio Optimization with Proportional Transaction Costs, *Mathematical Finance* 16, 301–335.
- Myers, Jeremy D., 2009, Portfolio Optimization with Transaction Costs and Preconceived Portfolio Weights, Ph.d dissertation, Massachusetts Institute of Technology.
- Olivares-Nadal, Alba V., and Victor DeMiguel, 2016, A Robust Perspective on Transaction Costs in Portfolio Optimization, *Operation Research* 66, 733–739.
- Pezzo, Luca, Lei Wang, and Duygu Zirek, 2020, Mean-variance market timing the U.S. Stock Market, Working paper, University of New Orleans and Montclair State University.
- Pogue, G. A., 1970, An Extension of the Markowitz Portfolio Selection Problem to Include Variable Transactions' Costs, Short Sales, Leverage Policies and Taxes, *Journal of Finance* 25, 1005–1027.
- Rockafellar, R. Tyrrell, 1970, Convex analysis, Technical report, Princeton University Press.
- Schachermayer, Walter, 2017, Asymptotic Theory of Transaction Costs , in Jessica James, Ian W. Marsh, and Lucio Sarno, eds., *Asymptotic Theory of Transaction Costs* (European Mathematical Society, Hoboken, NJ).
- Shreve, Steven E., and Halil M. Soner, 1994, Optimal Investment and Consumption with Transaction Costs, *The Annals of Applied Probability* 4, 609–692.
- Taksar, M., M. J. Klass, and D. Assaf, 1988, A Diffusion Model for Optimal Portfolio Selection in the Presence of Brokerage Fees, *Mathematics of Operations Research* 13, 277–294.
- Tobin, J., 1958, Liquidity Preference as Behavior Towards Risk, *Review of Economic Studies* 25, 65–86.

Utility loss from transaction costs

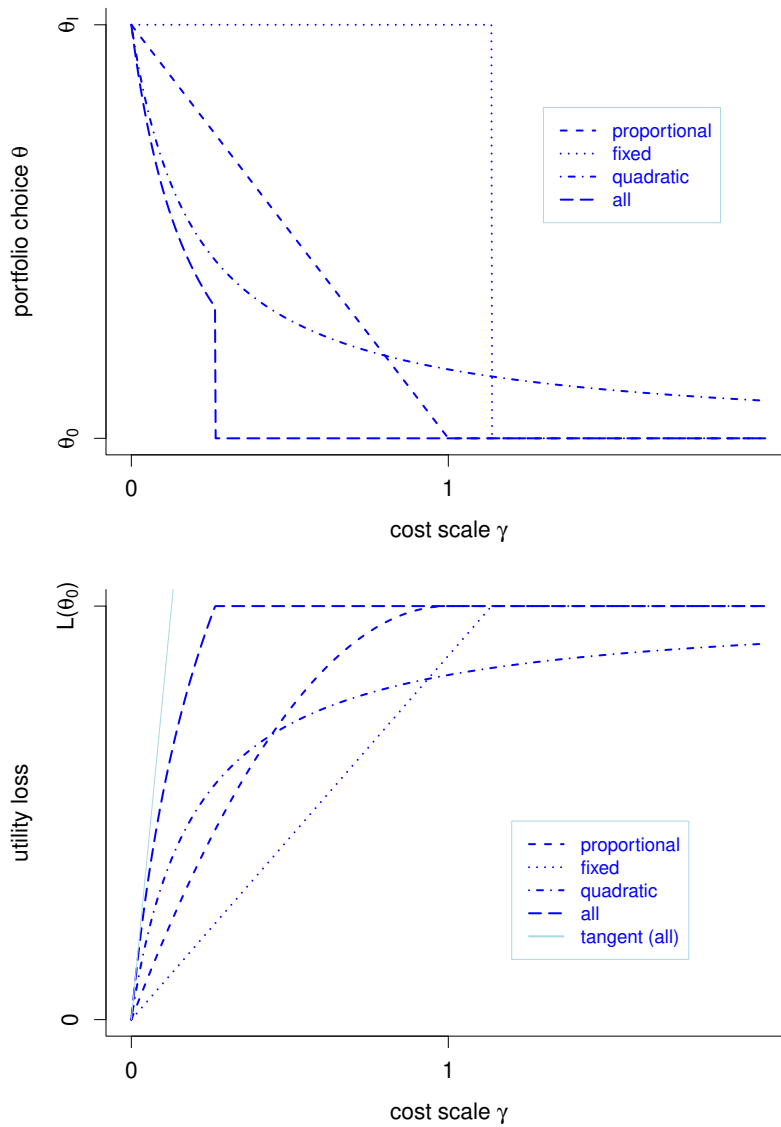


Figure 1: If the initial portfolio θ_0 is not the ideal portfolio θ_I that would be optimal absent costs, the impact of transaction costs is approximately proportional for small enough costs and approximately constant for large enough costs. This is true whether the costs are fixed, proportional, quadratic, or a combination, as is illustrated in the single-risky-security example in the figure. When costs are small, we trade approximately to the ideal portfolio θ_I , and the utility loss is approximately equal to the loss from trading to θ_I , which is proportional to the cost, as in the light blue line for the case of all costs. In contrast, when costs are large, we trade very little and the utility loss is approximately equal to the utility loss $L(\theta_0)$ of staying at θ_0 and not trading at all. The objective function (all costs) is $\theta'\mu - \lambda\theta'V\theta/2 - \kappa(\theta - \theta_B)'V(\theta - \theta_B)/2 - \gamma(C_+(\theta - \theta_0)^+ + C_-(\theta_0 - \theta)^+ + K(\{n|(\theta - \theta_0)_n \neq 0\}) + (\theta - \theta_0)'\Pi(\theta - \theta_0)/2)$, where $\mu = 0.05$, $\lambda = 2$, $V = 0.04$, $\kappa = 0$, arbitrary θ_B , $\theta_0 = 0.5$, $C_+ = C_- = 0.01$, $K(\emptyset) = 0$, $K(\{1\}) = 0.00055$, and $\Pi = 0.4$.

Mean-variance problem with proportional costs

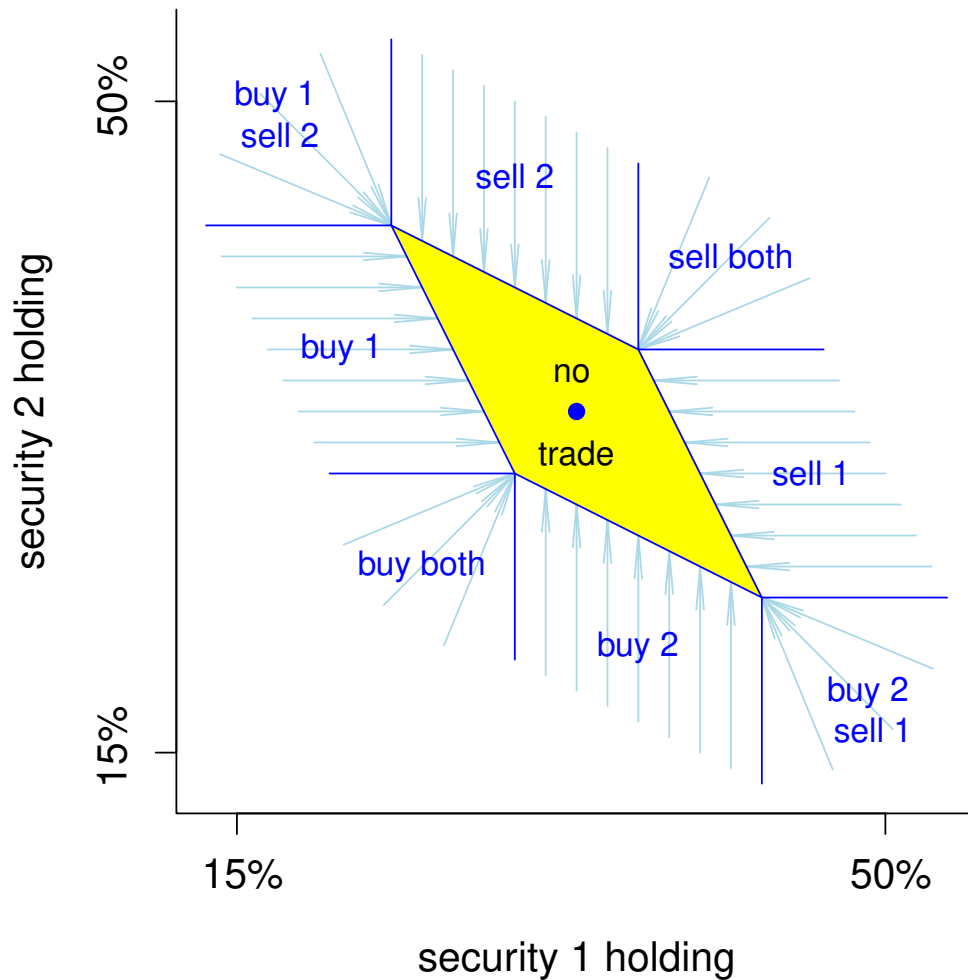


Figure 2: With only proportional costs, the marginal trading cost in some direction is constant. Hence any trade goes to the boundary of the nontrading region (NTR) but no further. In this example with two symmetric positively correlated risky securities, the NTR is the yellow parallelogram (which would be a rectangle absent correlation), and the ideal portfolio absent costs is the blue dot in the center. Outside the NTR, when costs are only proportional, it is optimal to trade as shown by the arrows. The positive correlation implies that securities are substitutes and we are quicker to trade when both securities are out of line in the same direction. If we added quadratic costs (perhaps from price pressure), the NTR would be the same because the marginal quadratic cost is zero at the starting portfolio, but any trading from outside the NTR (not shown) would stop short of the NTR due to the increasing marginal cost. The figure's objective function is $\theta' \mu - \lambda \theta' V \theta / 2 - \kappa (\theta - \theta_B)' V (\theta - \theta_B) / 2 - C_+(\theta - \theta_0)^+ - C_-(\theta_0 - \theta)^+$, where $\mu' = [0.02, 0.02]'$, $V_{11} = V_{22} = 0.04$, $V_{12} = V_{21} = 0.02$, $C'_+ = C'_- = [0.0002, 0.0002]'$, $\kappa = 0$, arbitrary θ_B , and $\lambda = 1$.

Traditional futures overlay with proportional costs

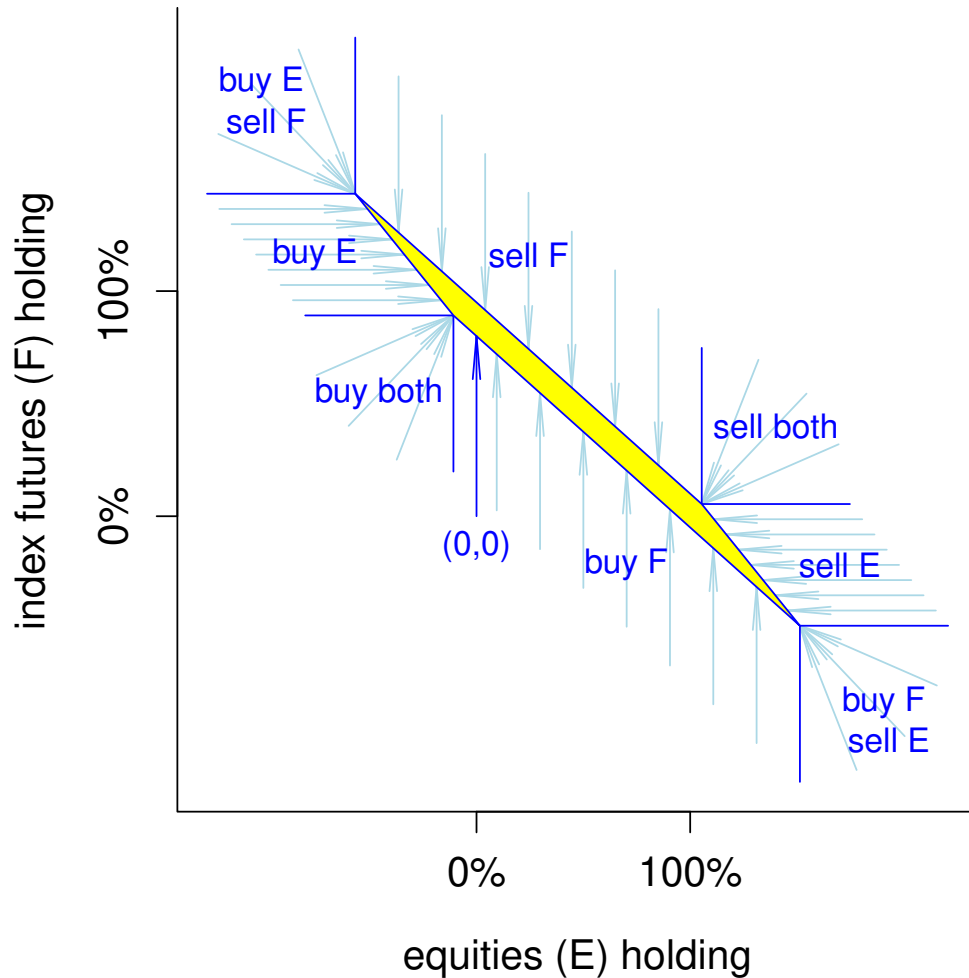


Figure 3: A futures overlay strategy manages transaction costs by trading a highly correlated equity futures contract with low trading costs in place of trading equities. Traditionally, futures overlays have been symmetric, trading futures to address either over-weighting or under-weighting in equity exposure, as would be optimal in this example. Starting from the normal range of equities positions (0% through 100%) it is optimal to pursue a traditional futures overlay strategy, correcting under- or over-exposure to market risk by trading futures. Unfortunately, while a symmetric overlay is optimal in this example, there is no reason ever to hold equities if we start from all cash ((0,0) in the graph), and instead we should follow a pure synthetic equity strategy. Because equities and futures carry the same risk premium, the cheaper trading cost of futures is decisive. The objective function is as in Figure 2, with parameters $\mu' = [0.0175, 0.0175]'$, $V_{11} = V_{22} = 0.04$, $V_{12} = V_{21} = 0.034$, $C'_+ = C'_- = [0.0045, 0.0015]'$, $\kappa = 0$, arbitrary θ_B , and $\lambda = 0.5$.

Realistic futures overlay with proportional costs

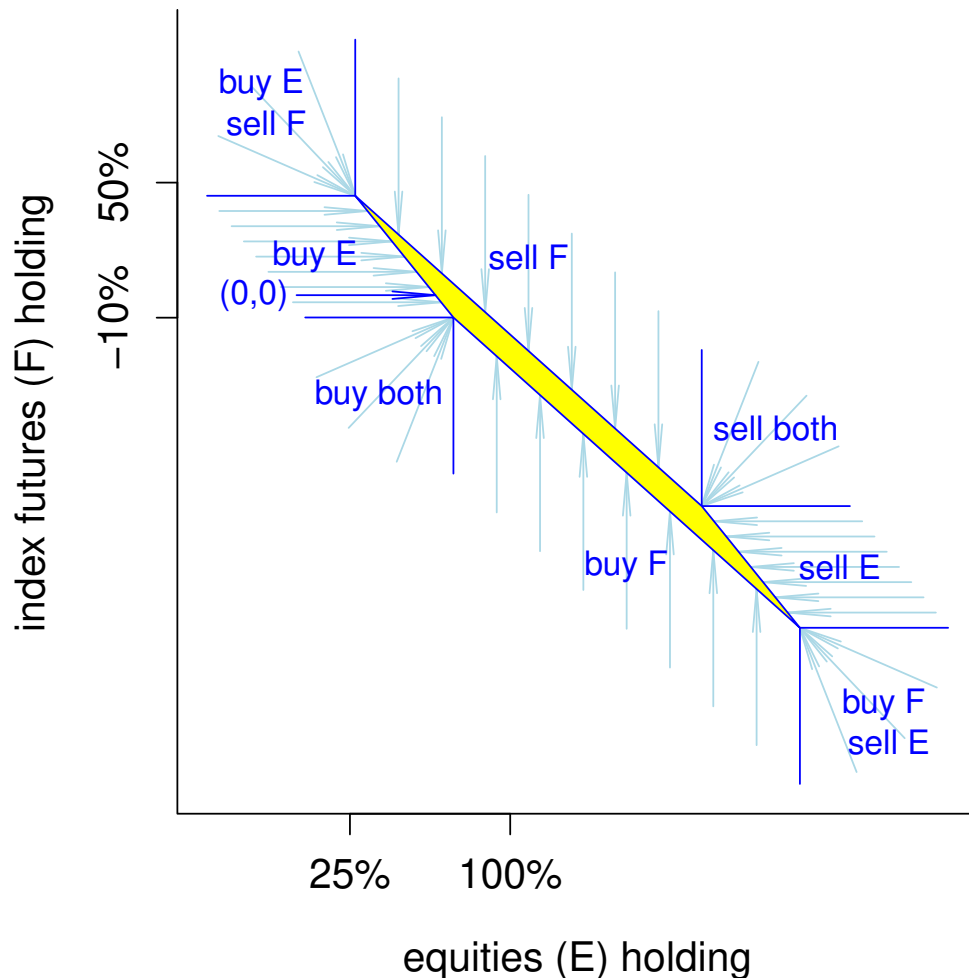


Figure 4: If we assume equities have a higher risk premium than equity futures, perhaps because the equities are actively managed, we find it is optimal to undertake an asymmetric futures overlay strategy of buying equities and selling futures in the normal range (25% to 100% equities and -10% to 50% futures). The larger risk premium on equities makes it optimal to buy equities starting from all cash $(0,0)$, and also makes it optimal to buy equities to increase market exposure but sell futures to reduce effective equity exposure. In both directions, we make the choice that has larger alpha. The objective function and parameters are the same as in Figure 3 except with unequal means $\mu' = [0.0175, 0.012]'$.

Mean-variance problem with an overall fixed cost

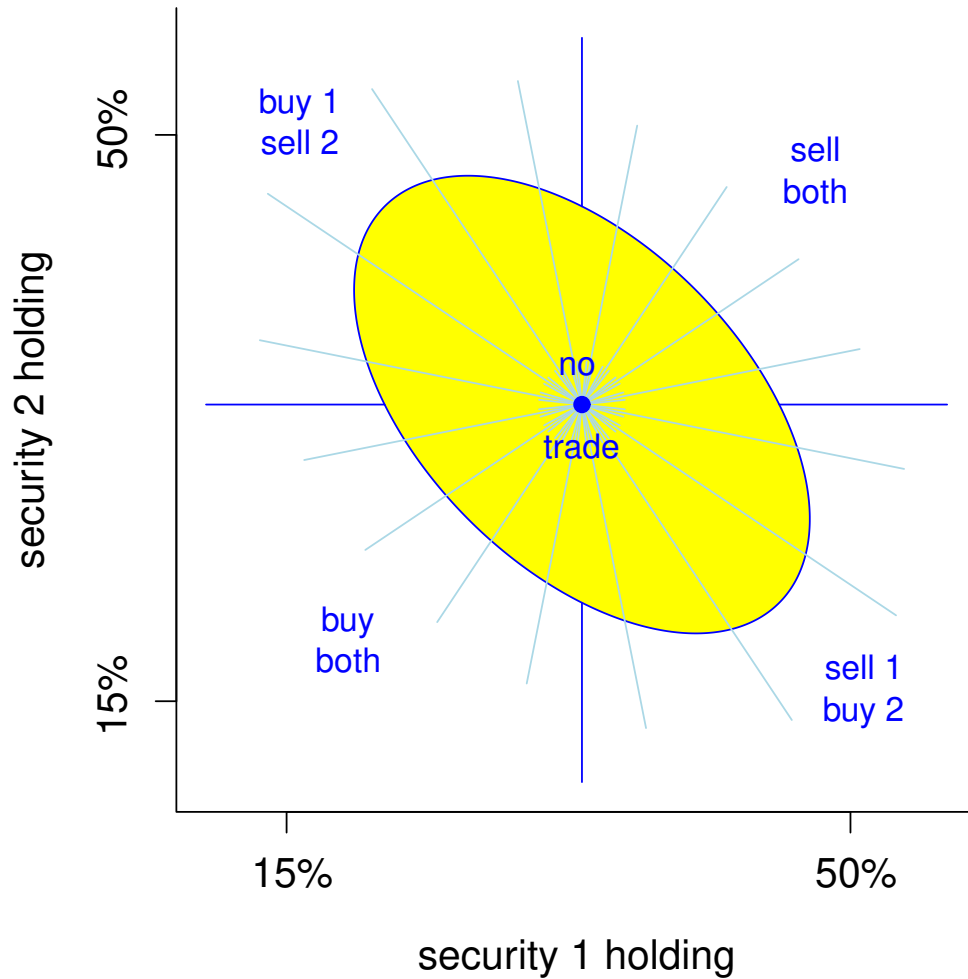


Figure 5: With only an overall fixed cost, if it is optimal to trade at all, we trade to the ideal portfolio, because the cost is sunk once it is triggered. The marginal cost of trading is a limiting case of decreasing marginal cost (infinite initially and constant thereafter), hence any optimal trade goes inside the nontrading region (NTR). In this example with two symmetric positively correlated risky securities, the NTR is the shaded yellow ellipse and all trades go to the ideal portfolio, the blue dot. As with proportional costs, positive correlation between the securities implies the assets are substitutes, so that the value of trading overcomes the cost sooner when both asset positions are out of line in the same direction. Adding quadratic costs (perhaps due to price pressure) would not change the qualitative shape of the NTR but would make it larger. With both overall fixed and quadratic costs, the overall marginal cost of trading would not be monotonic and consequently the optimal trades (not shown) might land inside, outside or on the boundary of the NTR, depending on how far from the NTR we start. The figure's objective function is $\theta' \mu - \lambda \theta' V \theta / 2 - \kappa (\theta - \theta_B)' V (\theta - \theta_B) / 2 - K(\{n | (\theta - \theta_0)_n \neq 0\})$, where $\mu' = [0.02, 0.02]'$, $V_{11} = V_{22} = 0.04$, $V_{12} = V_{21} = 0.02$, $\kappa = 0$, θ_B is arbitrary, $\lambda = 1$, $K(S) = 0.0003$ for $S \neq \emptyset$, and $K(\emptyset) = 0$.

Mean-variance problem with security-specific fixed costs

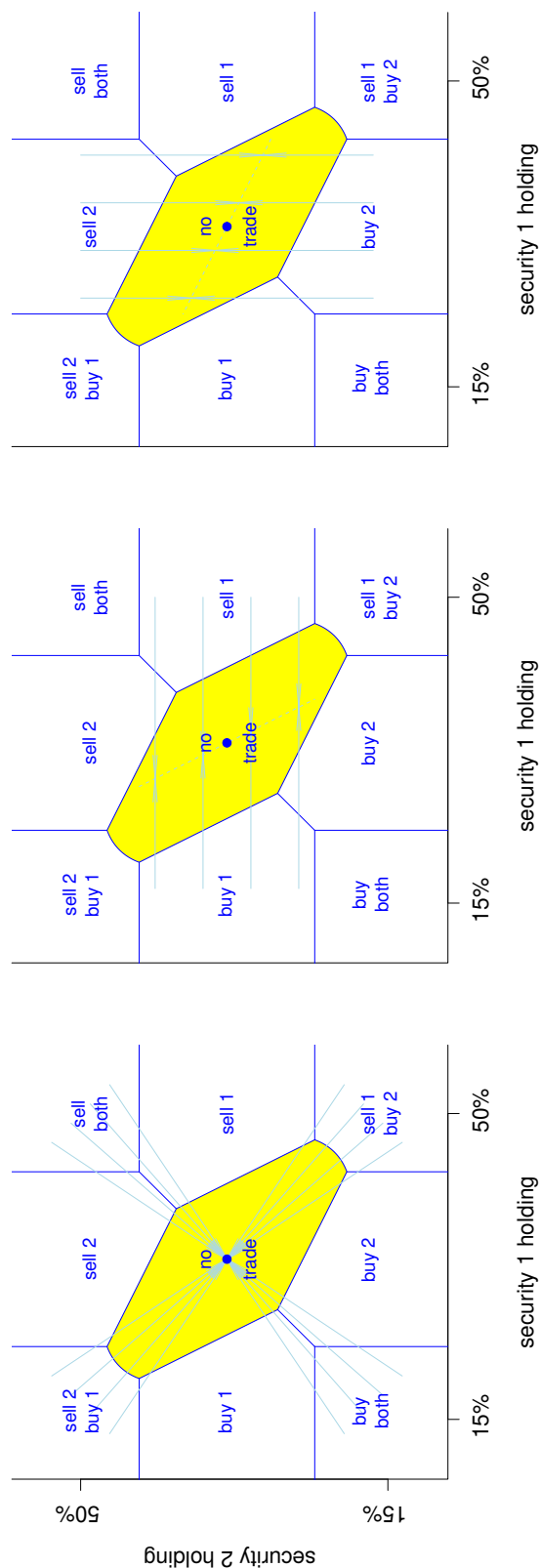


Figure 6: With security-specific costs, it only pays to trade securities when they are significantly out of line with the ideal allocation (the blue dot). Similarly to the overall fixed cost case, the marginal cost of trading in some direction that has a fixed cost is a limiting case of decreasing marginal cost (infinite at first and constant thereafter). Hence, any (nonzero) optimal trade goes inside the NTR, but not necessarily to the ideal point. This example has two symmetric positively correlated risky securities. From left to right, the arrows indicate the optimal trades (i) to the ideal portfolio when both securities are traded, (ii) to a line inside the NTR going through the optimal portfolio when only security 1 is traded, and (iii) to another internal line passing through the ideal portfolio when only security 2 is traded. If we are trading only one security, how much we trade is decreasing in our position in the other security because positive correlation makes the securities substitutes. The objective function is $\theta'\mu - \lambda\theta'V\theta/2 - \kappa(\theta - \theta_B)'V(\theta - \theta_B)/2 - K(\{n|\theta_n \neq \theta_n^0\})$, where $\mu' = [0.02, 0.02]'$, $V_{11} = V_{22} = 0.04$, $V_{12} = V_{21} = 0.02$, $\kappa = 0$, θ_B is arbitrary, $\lambda = 1$ and $K(\{1\}) = K(\{2\}) = 0.00015$, $K(\{1, 2\}) = 0.0003$, and $K(\emptyset) = 0$.

Mean-variance problem with proportional and asset-specific fixed costs

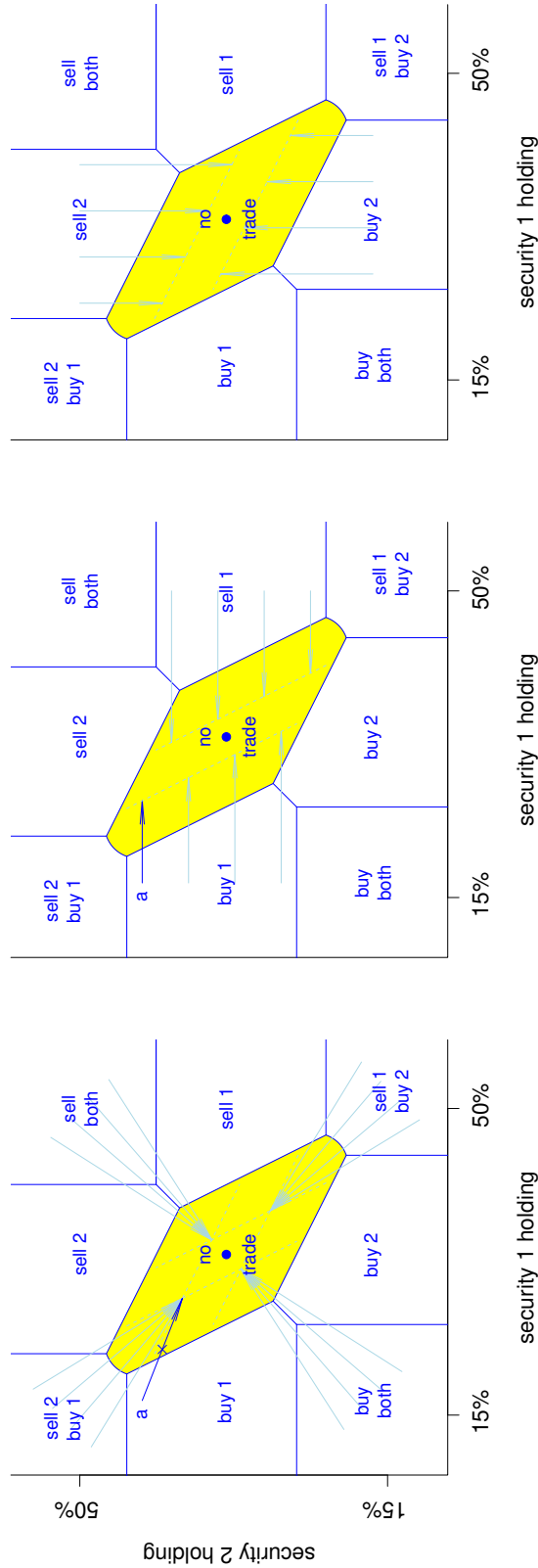


Figure 7: With both proportional and security-specific costs, the optimal trade is the same as the optimal trade with only proportional costs when the starting portfolio is far enough from the ideal portfolio for all securities, as in the four corners of the left panel. In this example, the inside parallelogram would be the NTR absent the fixed costs (as in Figure 2). Once it is obvious it will be optimal to trade something and incur the fixed cost, we can view the fixed cost as sunk. Furthermore, if we are far away from the ideal position for only some securities, those securities' fixed costs of trading can be considered as sunk. However, the same may not be true for the rest of the securities. For example, if $\theta_0 = a$, we would sell 1 and buy 2 (first panel, crossed out) if we did not have a fixed cost of trading security 2 (as in Figure 2), but with security-specific fixed costs it is not worth trading security 2 (middle panel). The objective function is $\theta' \mu - \lambda \theta' V \theta / 2 - \kappa(\theta - \theta_B)' V(\theta - \theta_B) / 2 - K(\{\eta | \theta_n \neq \theta_n^0\}) - C_+(\theta - \theta_0)^+ - C_-(\theta - \theta_0)^-$, where $\mu' = [0.02, 0.02]'$, $V_{11} = V_{22} = 0.04$, $V_{12} = V_{21} = 0.02$, $C'_+ = C'_- = [0.0002, 0.0002]'$, $\kappa = 0$, arbitrary θ_B , $\lambda = 1$, $K(\{1\}) = K(\{2\}) = 0.00015$, $K(\{1, 2\}) = 0.0003$, and $K(\emptyset) = 0$.